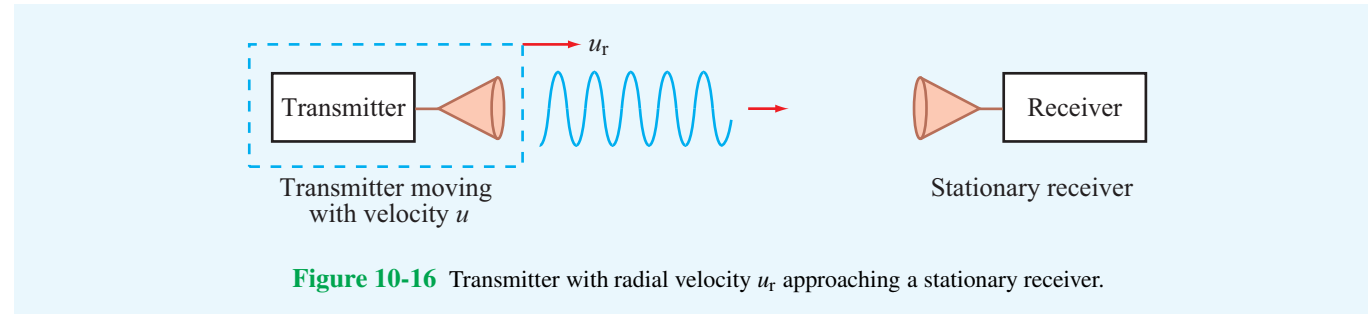




with frequency f_t (Fig. 10-16). At a distance R from the source, the electric field of the radiated wave is given by

$$E(R) = E_0 e^{j(\omega_t t - kR)} = E_0 e^{j\phi}, \quad (10.29)$$

where E_0 is the wave's magnitude, $\omega_t = 2\pi f_t$, and $k = 2\pi/\lambda_t$ with λ_t being the wavelength of the transmitted wave. The magnitude depends on the distance R and the gain of the source antenna, but it is not of concern as far as the Doppler effect is concerned. The quantity

$$\phi = \omega_t t - kR = 2\pi f_t t - \frac{2\pi}{\lambda_t} R \quad (10.30)$$

is the phase of the radiated wave relative to its phase at $R = 0$ and reference time $t = 0$. If the source is moving toward the receiver, as in Fig. 10-16, or vice versa, at a radial velocity u_r , then

$$R = R_0 - u_r t, \quad (10.31)$$

where R_0 is the distance between the source and the receiver at $t = 0$. Hence,

$$\phi = 2\pi f_t t - \frac{2\pi}{\lambda_t} (R_0 - u_r t). \quad (10.32)$$

This is the phase of the signal detected by the receiver. The frequency of a wave is defined as the time derivative of the phase ϕ divided by 2π . Thus,

$$f_r = \frac{1}{2\pi} \frac{d\phi}{dt} = f_t + \frac{u_r}{\lambda_t}. \quad (10.33)$$

Comparison of Eq. (10.33) with Eq. (10.28) leads to $f_d = u_r/\lambda_t$. For radar, the Doppler shift happens twice: once for the wave from the radar to the target and again for the wave reflected by the target back to the radar. Hence, $f_d = 2u_r/\lambda_t$. The dependence of f_d on direction is given by the dot product of the velocity and range unit vectors, which leads to

$$f_d = -2 \frac{u_r}{\lambda_t} = -\frac{2u}{\lambda_t} \cos \theta, \quad (10.34)$$

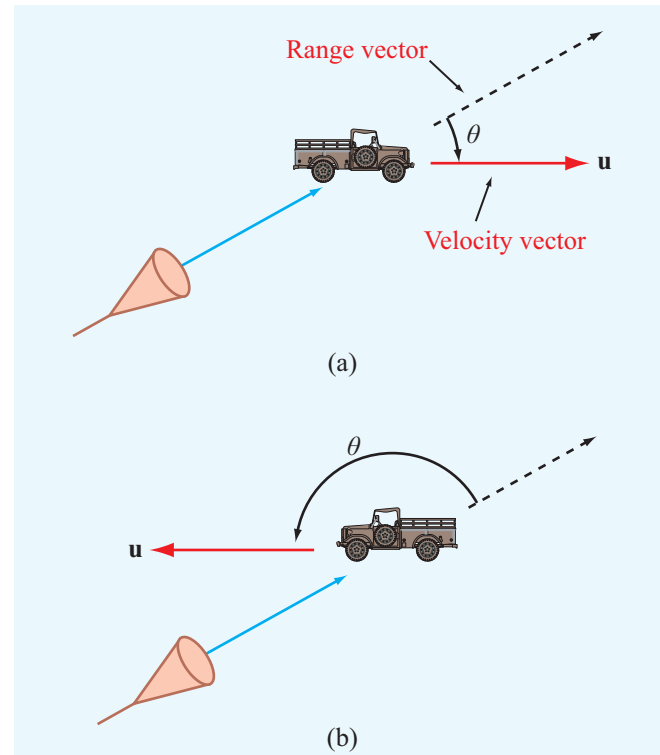


Figure 10-17 The Doppler frequency shift is negative for a receding target ($0 \leq \theta \leq 90^\circ$), as in (a), and positive for an approaching target ($90^\circ \leq \theta \leq 180^\circ$), as in (b).

where u_r is the radial velocity component of u and θ is the angle between the range vector and the velocity vector (Fig. 10-17) with the direction of the range vector defined to be from the radar to the target. For a receding target (relative to the radar), $0 \leq \theta \leq 90^\circ$, and for an approaching target, $90^\circ \leq \theta \leq 180^\circ$.

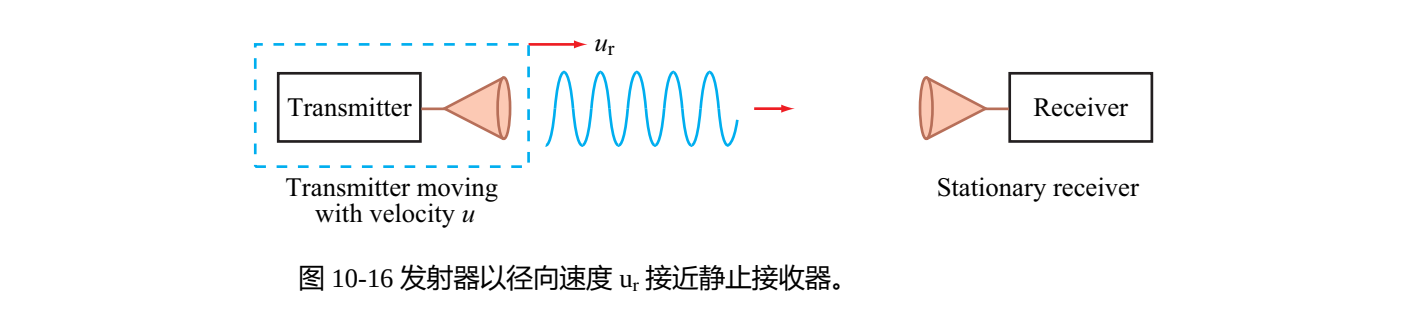


图 10-16 发射器以径向速度 u_r 接近静止接收器。

频率为 f_t (10-16)。在距源的距离 R 处，辐射波的电场由下式给出

$$E(R) = E_0 e^{j(\omega_t t - kR)} = E_0 e^{j\phi}, \quad (10.29)$$

其中 E_0 是波的幅度， $\omega_t = 2\pi f_t$ 和 $k = 2\pi/\lambda_t$ ，其中 λ_t 是传输波的波长。幅度取决于距离 R 和源天线的增益，但就多普勒效应而言，它并不重要。数量

$$\phi = \omega_t t - kR = 2\pi f_t t - \frac{2\pi}{\lambda_t} R \quad (10.30)$$

是辐射波的相位相对于其在 $R = 0$ 和参考时间 $t = 0$ 时的相位。如果源以径向速度 u_r 向接收器移动，如图 10-16 所示，或反之亦然，则

$$R = R_0 - u_r t, \quad (10.31)$$

其中 R_0 是 $t = 0$ 时源与接收器之间的距离。因此，

$$\phi = 2\pi f_t t - \frac{2\pi}{\lambda_t} (R_0 - u_r t). \quad (10.32)$$

这是接收器检测到的信号的相位。波的频率定义为相位 ϕ 除以 2π 的时间导数。因此，

$$f_r = \frac{1}{2\pi} \frac{d\phi}{dt} = f_t + \frac{u_r}{\lambda_t}. \quad (10.33)$$

等式的比较 (10.33) 与方程 (10.28) 导致 $f_d = u_r/\lambda_t$ 。对于雷达，多普勒频移发生两次：一次是波从雷达到目标，另一次是波被目标反射回雷达。因此， $f_d = 2u_r/\lambda_t$ 。 f_d 对方向的依赖性由速度和距离单位向量的点积给出，这导致

$$f_d = -2 \frac{u_r}{\lambda_t} = -\frac{2u}{\lambda_t} \cos \theta, \quad (10.34)$$

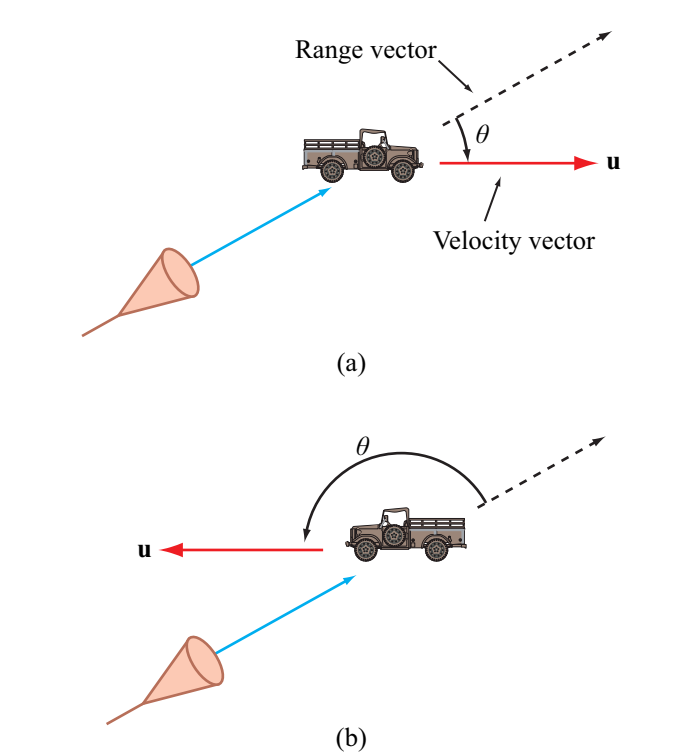


图 10-17 后退目标的多普勒频移为负 ($0 \leq \theta \leq 90^\circ$)，如 (a) 所示；对于接近的目标，多普勒频移为正 ($90^\circ \leq \theta \leq 180^\circ$)，如 (b) 所示。

其中 u_r 是 u 的径向速度分量， θ 是距离矢量和速度矢量之间的角度 (图 10-17)，距离矢量的方向定义为从雷达到目标。对于后退目标 (相对于雷达)， $0 \leq \theta \leq 90^\circ$ ，对于接近目标， $90^\circ \leq \theta \leq 180^\circ$ 。

10-8 Monopulse Radar

On the basis of information extracted from the echo due to a single pulse, a **monopulse radar** can track the direction of a target with an angular accuracy equal to a fraction of its antenna beamwidth. To track a target in both elevation and azimuth, a monopulse radar uses an antenna (such as a parabolic dish) with four separate small horns at its focal point (Fig. 10-18). Monopulse systems are of two types. The first is called **amplitude-comparison monopulse** because the tracking information is extracted from the amplitudes of the echoes received by the four horns, and the second is called **phase-comparison monopulse** because it relies on the phases of the received signals. We shall limit our present discussion to the amplitude-comparison scheme.

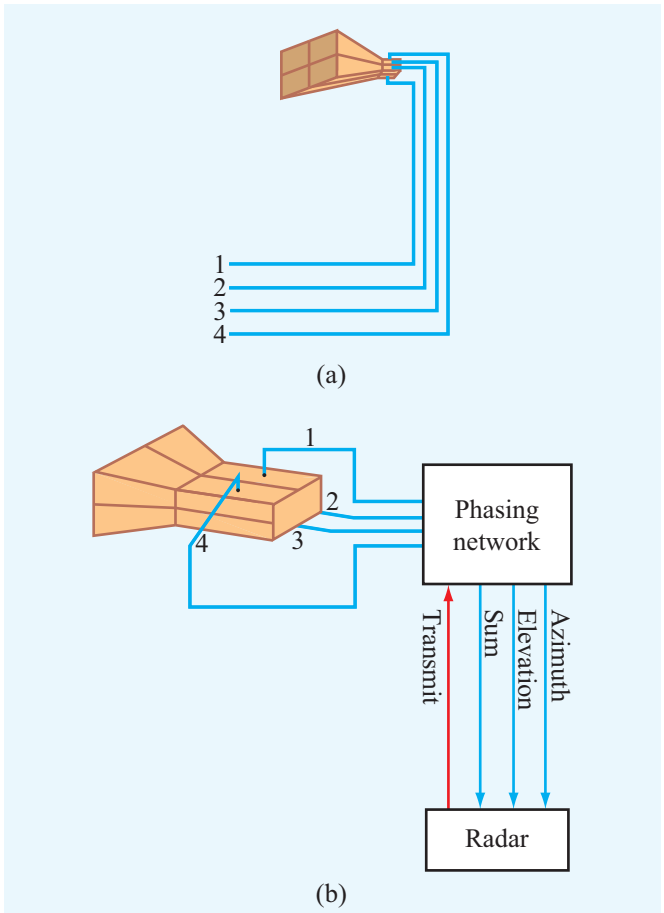


Figure 10-18 Antenna feeding arrangement for an amplitude-comparison monopulse radar: (a) feed horns and (b) connection to phasing network.

Individually, each horn would produce its own beam with the four beams pointing in slightly different directions. Figure 10-19 shows the beams of two adjacent horns. The basic principle of the amplitude-comparison monopulse is to measure the amplitudes of the echo signals received through the two beams and then apply the difference between them to repoint the antenna boresight direction toward the target. Using computer-controlled phase shifters, the **phasing network** shown in Fig. 10-18 can combine the signal delivered to the four-element horn array by the transmitter or by the echo signals received by them in different ways. Upon transmission, the network excites all four feeds in phase, thereby producing a single main beam called the **sum beam**. The phasing network uses special microwave devices that allow it to provide the desired functionality during both the transmit and receive modes. Its equivalent functionality is described by the circuits shown in Fig. 10-20. During the receive period, the phasing network uses power dividers, power combiners, and phase shifters to generate three different output channels. One of these is the sum channel, corresponding to adding all four horns in phase, and its radiation pattern is depicted in Fig. 10-21(a). The second channel, called the **elevation-difference channel**, is obtained by first adding the outputs of the top-right and top-left horns (Fig. 10-20(b)), then adding the outputs of the bottom-right and bottom-left horns, and finally subtracting the second sum from the first. The subtraction process is accomplished by adding a 180° phase shifter in the path of the second sum before adding it to the first sum. The beam pattern of the elevation-difference channel is shown in Fig. 10-21(b). If the observed target is centered between the two elevation beams, the receiver echoes will have the same strength for both beams, thereby producing a zero output from the elevation-difference channel. If it is not, the amplitude of the elevation-difference channel will be proportional to the angular deviation of the target from the boresight direction, and its sign will denote the direction of the deviation. The third channel (not shown in Fig. 10-20) is the **azimuth-difference channel**, and it is

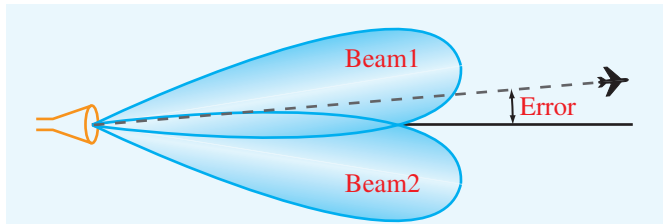


Figure 10-19 A target observed by two overlapping beams of a monopulse radar.

10-8 单脉冲雷达

基于从单个脉冲的回波中提取的信息，单脉冲雷达可以跟踪目标的方向，其角度精度等于其天线波束宽度的一部分。为了在仰角和方位角上跟踪目标，单脉冲雷达使用在其焦点处带有四个独立小喇叭的天线（例如抛物面天线）（图10-18）。单脉冲系统有两种类型。第一种称为幅度比较单脉冲，因为跟踪信息是从四个喇叭接收到的回波的幅度中提取的；第二种称为相位比较单脉冲，因为它依赖于接收信号的相位。我们将把目前的讨论限制在幅度比较方案上。

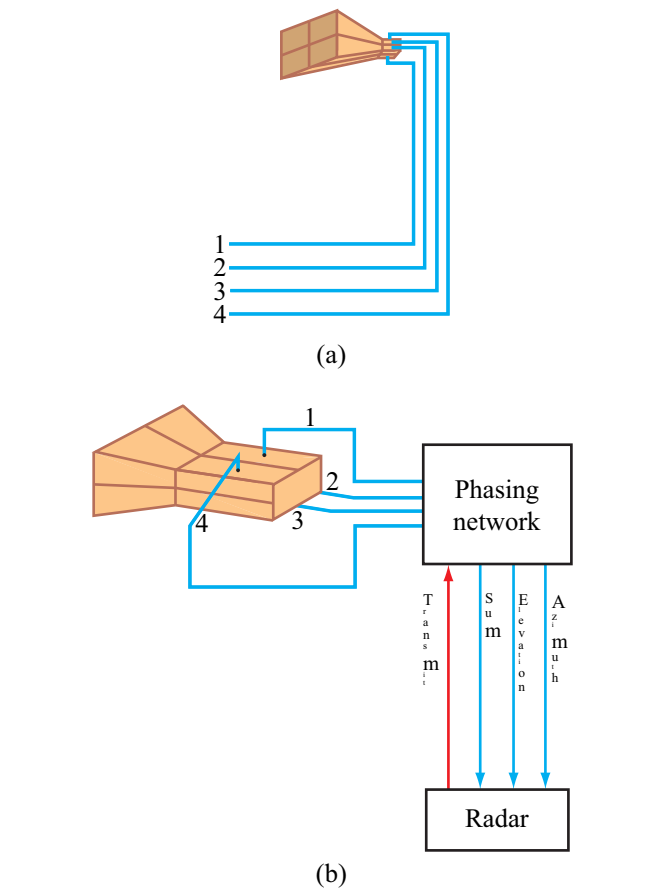


图 10-18 幅度比较单脉冲雷达的天线馈电布置：(a) 馈电喇叭和 (b) 与定相网络的连接。

每个喇叭都会单独产生自己的波束，四个波束指向略有不同的方向。图10-19显示了两个相邻喇叭的波束。幅度比较单脉冲的基本原理是测量通过两个波束接收到的回波信号的幅度，然后应用它们之间的差值将天线视轴方向重新指向目标。使用计算机控制的移相器，图10-18所示的定相网络可以将发射器传递到四元件喇叭阵列的信号或它们以不同方式接收到的回波信号组合起来。传输时，网络同相激励所有四个馈源，从而产生称为和波束的单个主波束。定相网络使用特殊的微波设备，使其能够在发射和接收模式下提供所需的功能。其等效功能由图 10-20 所示的电路描述。在接收期间，定相网络使用功率分配器、功率组合器和移相器来生成三个不同的输出通道。其中之一是求和通道，相当于将所有四个喇叭同相相加，其辐射图如图 10-21(a) 所示。第二个通道称为俯仰差通道，首先将右上角和左上角的输出相加（图 10-20 (b) ），然后将右下角和左下角的输出相加，最后从第一个和中减去第二个和。减法过程是通过在将第二个和添加到第一个和之前在第二个和的路径中添加一个 180° 移相器来完成的。俯仰差通道的波束方向图如图10-21(b) 所示。如果观察到的目标位于两个仰角波束之间的中心，则两个波束的接收器回波将具有相同的强度，从而从仰角差通道产生零输出。如果不是，则仰角通道的幅度将与目标相对视轴方向的角度偏差成正比，其符号表示偏差的方向。第三个通道（图10-20中未示出）是方位差通道，它是

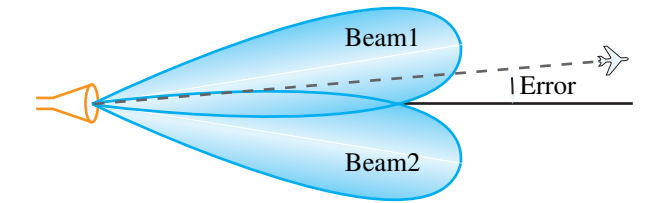


图10-19 单脉冲雷达的两个重叠波束观测到的目标。

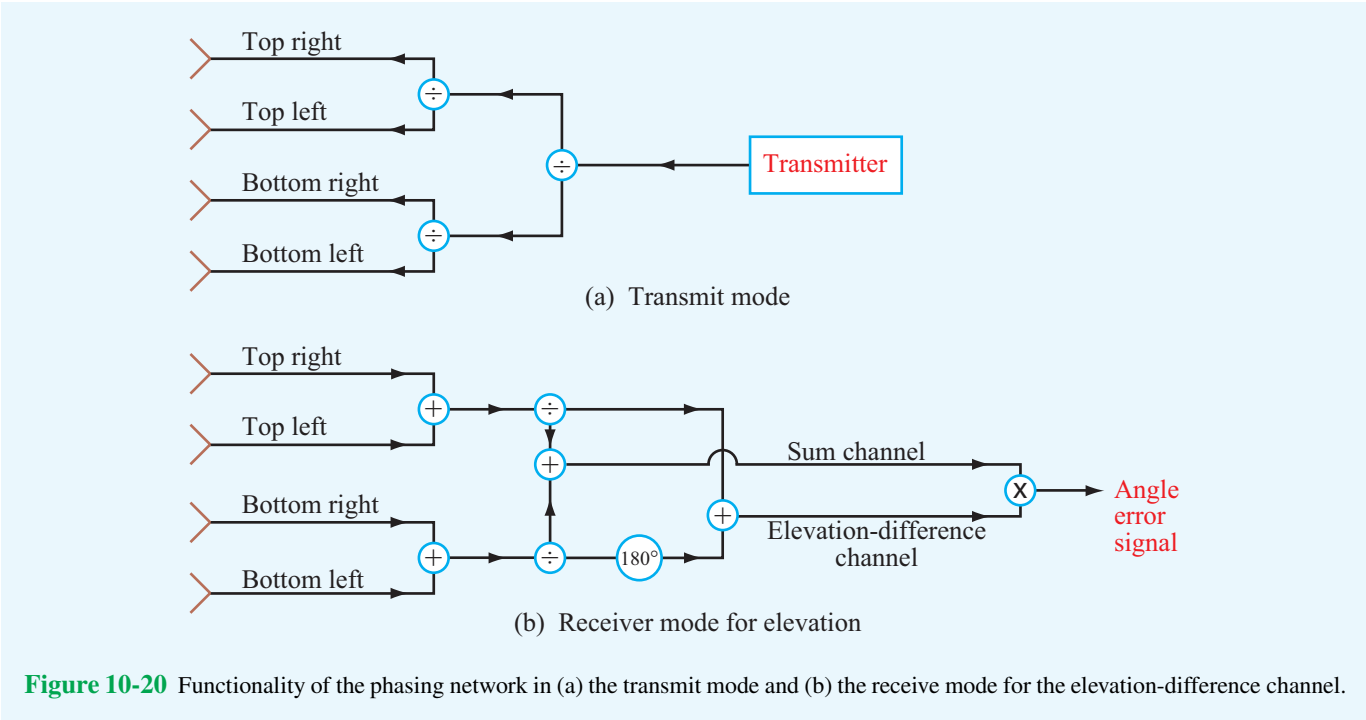


Figure 10-20 Functionality of the phasing network in (a) the transmit mode and (b) the receive mode for the elevation-difference channel.

accomplished through a similar process that generates a beam corresponding to the difference between the sum of the two right horns and the sum of the two left horns.

In practice, the output of the difference channel is multiplied by the output of the sum channel to increase the strength of the difference signal and to provide a phase reference for extracting the sign of the angle. This product, called the **angle error signal**, is displayed in Fig. 10-21(c) as a function of the angle error. The error signal activates a servo-control system to reposition the antenna direction. By applying a similar procedure along the azimuth direction using the product of the azimuth-difference channel and the sum channel, a monopulse radar provides automatic tracking in both directions. The range to the target is obtained by measuring the round-trip delay of the signal.

Concept Question 10-5: How is the PRF related to unambiguous range?

Concept Question 10-6: Explain how the false-alarm probability and the detection probability are related to the noise level of the receiver.

Concept Question 10-7: In terms of the geometry shown in Fig. 10-17, when is the Doppler shift a maximum?

Concept Question 10-8: What is the principle of the monopulse radar?

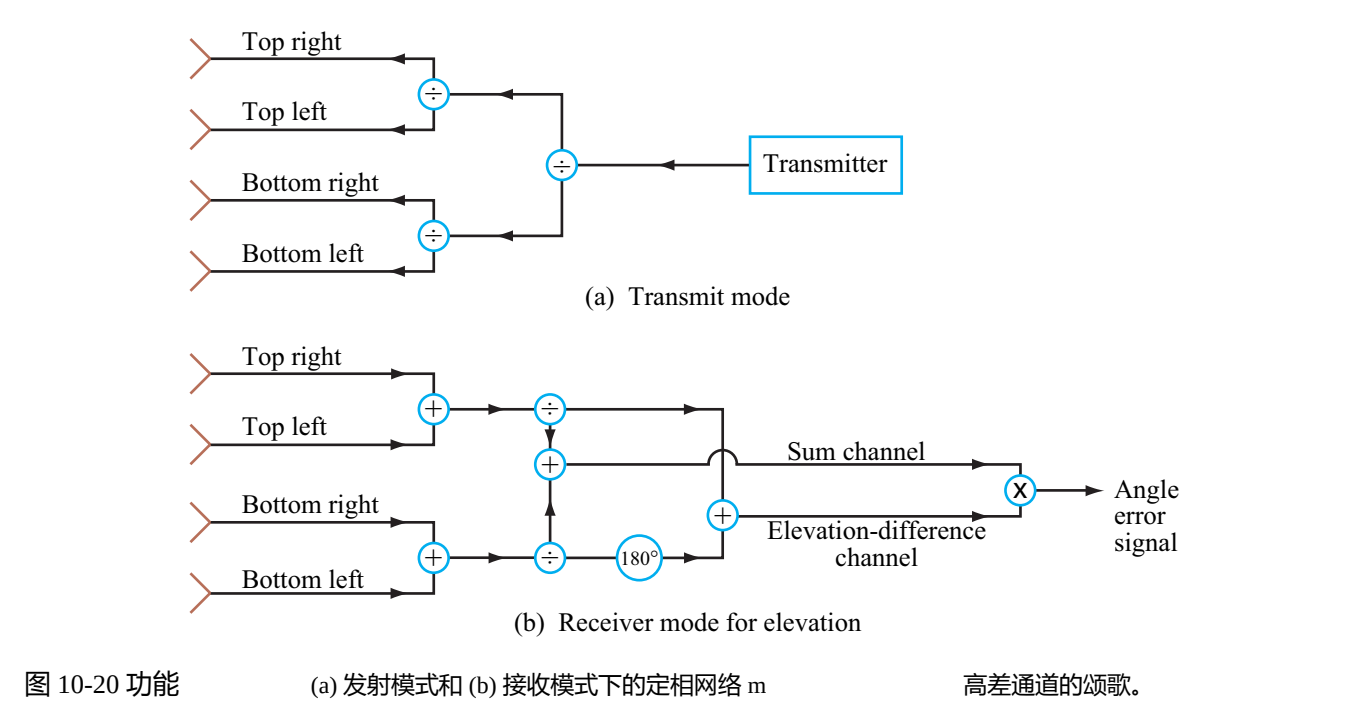


图 10-20 功能 (a) 发射模式和 (b) 接收模式下的定相网络 m 高差通道的颂歌。

通过类似的过程来完成，该过程生成与两个右喇叭的总和与两个左喇叭的总和之间的差相对应的光束。

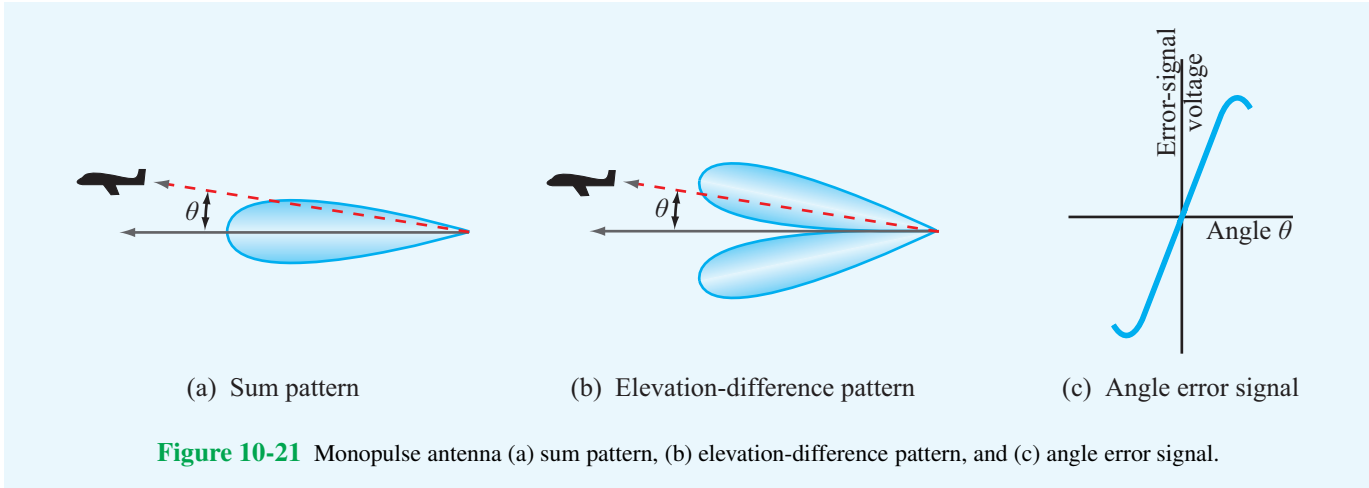
实际上，差通道的输出乘以和通道的输出，以增加差信号的强度并为提取角度符号提供相位参考。该乘积称为角度误差信号，作为角度误差的函数显示在图 10-21(c) 中。误差信号激活伺服控制系统以重新定位天线方向。通过使用方位差通道与和通道的乘积沿着方位角方向应用类似的过程，单脉冲雷达提供两个方向的自动跟踪。通过测量信号的往返延迟来获得目标的距离。

概念问题 10-5: PRF 与明确范围有何关系?

概念问题 10-6: 解释误报概率和检测概率与接收器噪声水平的关系。

概念问题10-7: 根据图10-17所示的几何形状，多普勒频移何时达到最大值?

概念题10-8: 单脉冲雷达的原理是什么?



Chapter 10 Summary

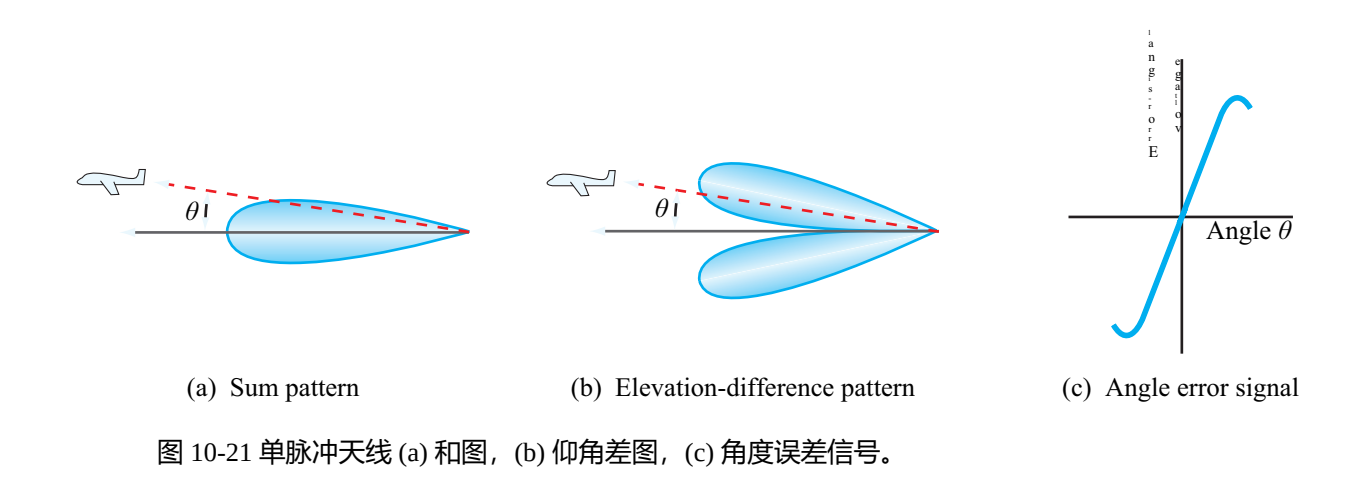
Concepts

- Three equally spaced satellites in geostationary orbit can provide coverage of most of Earth’s surface.
- The use of polarization diversity makes it possible to double the number of channels per unit bandwidth carried by a satellite repeater.
- A satellite antenna system is designed to produce beams tailored to match the areas served by the satellite. Antenna arrays are particularly suitable for this purpose.
- A radar is an electromagnetic sensor that illuminates a region of space and then measures the echoes due to reflecting objects. From the echoes, information can be extracted about the range of a target, its radial velocity, direction of motion, and other characteristics.
- Due to the random nature of receiver noise, target detection is a statistical process characterized by detection and false-alarm probabilities.
- A moving object produces a Doppler frequency shift proportional to the radial velocity of the object (relative to the radar) and inversely proportional to λ .
- A monopulse radar uses multiple beams to track the direction of a target with an angular accuracy equal to a fraction of its antenna beamwidth.

Important Terms

Provide definitions or explain the meaning of the following terms:

atmospheric transmissivity Υ	lidar	radial velocity u_r
azimuth resolution	matched filter	range resolution
bistatic radar	maximum detectable range $R_{\max}$	Score
circulator	monopulse radar	signal-to-noise ratio
detection probability	monostatic radar	Sputnik I
Doppler frequency shift f_d	multiplexer	sum and difference channels
duplexer	polarization diversity	synchronizer
Explorer I	pulse length τ	system noise temperature
false-alarm probability	pulse repetition frequency (PRF) f_p	threshold detection level
FDMA	radar	transponder
geostationary orbit	radar cross section σ_t	unambiguous range R_u
interpulse period T_p	radar equation	uplink and downlink



第10章总结

概念

- 地球静止轨道上的三颗等距卫星可以覆盖地球表面的大部分区域。
- 极化分集的使用使得卫星中继器所承载的每单位带宽的信道数量加倍成为可能。
- 卫星天线系统旨在产生适合卫星服务区域的波束。天线阵列特别适合此目的。
- 雷达是一种电磁传感器，可以照亮空间区域，然后测量反射物体产生的回波。从回波中可以得到信息。
- 提取目标的范围、其径向速度、运动方向和其他特征。
- 由于接收机噪声的随机性，目标检测是一个以检测和误报概率为特征的统计过程。
- 移动物体产生的多普勒频移与物体的径向速度（相对于雷达）成正比，与 λ 成反比。
- 单脉冲雷达使用多个波束来跟踪目标的方向，其角度精度等于其天线波束宽度的一部分。

重要术语 提供以下术语的定义或解释其含义：

大气透射率 Υ	方位分辨率	激光雷达匹配滤波器	最大可探测距离 $R_{\max}$	单脉冲雷达	单基地雷达	多普勒频移 f_d	双工器	E	脉冲长度 τ	脉冲重复频率 (PRF) f_p	雷达截面 σ_t	雷达方程
双基地雷达	循环器	探测概率	多普勒频移 f_d	FDMA	对地静止轨道脉冲间周期 T_p	探测概率	FDMA	对地静止轨道脉冲间周期 T_p	探测概率	FDMA	对地静止轨道脉冲间周期 T_p	探测概率

Mathematical and Physical Models

Satellite Communication Systems

Radius of geostationary orbit

$$R_0 = \left(\frac{GM_{\text{e}}T^2}{4\pi^2} \right)^{1/3}$$

Received power

$$P_{\text{ri}} = \Upsilon(\theta) \, P_{\text{r}} = \Upsilon(\theta) \, P_{\text{t}}G_{\text{t}}G_{\text{r}} \left(\frac{\lambda}{4\pi R} \right)^2$$

Noise power

$$P_{\text{ni}} = KT_{\text{sys}}B$$

Signal-to-noise ratio

$$S_{\text{n}} = \frac{P_{\text{ri}}}{P_{\text{ni}}} = \frac{\Upsilon(\theta) \, P_{\text{t}}G_{\text{t}}G_{\text{r}}}{KT_{\text{sys}}B} \left(\frac{\lambda}{4\pi R} \right)^2$$

Radar Sensors

Unambiguous range

$$R_{\text{u}} = \frac{cT_{\text{p}}}{2} = \frac{c}{2f_{\text{p}}}$$

Range resolution

$$\Delta R = R_2 - R_1 = c\tau/2$$

Azimuth resolution

$$\Delta x = \beta R$$

Radar equation

$$P_{\text{r}} = \frac{P_{\text{t}}G^2\lambda^2\sigma_{\text{t}}}{(4\pi)^3R^4}$$

Doppler frequency shift

$$f_{\text{d}} = -2\frac{u_{\text{r}}}{\lambda_{\text{t}}} = -\frac{2u}{\lambda_{\text{t}}} \cos \theta$$

PROBLEMS

Sections 10-1 to 10-4: Satellite Communication Systems

- ***10.1** A remote sensing satellite is in circular orbit around Earth at an altitude of 1,100 km above Earth’s surface. What is its orbital period?
- 10.2** A transponder with a bandwidth of 400 MHz uses polarization diversity. If the bandwidth allocated to transmit a single telephone channel is 4 kHz, how many telephone channels can be carried by the transponder?

- ***10.3** Repeat Problem 10.2 for TV channels with each requiring a bandwidth of 6 MHz.

10.4 A geostationary satellite is at a distance of 40,000 km from a ground receiving station. The satellite transmitting antenna is a circular aperture with a 1 m diameter, and the ground station uses a parabolic dish antenna with an effective diameter of 20 cm. If the satellite transmits 1 kW of power at 12 GHz and the ground receiver is characterized by a system noise temperature of 1,000 K, what would be the signal-to-noise ratio of a received TV signal with a bandwidth of 6 MHz? The antennas and the atmosphere may be assumed lossless.

Sections 10-5 to 10-8: Radar Sensors

- ***10.5** A collision-avoidance automotive radar is designed to detect the presence of vehicles up to a range of 0.5 km. What is the maximum usable PRF?
- 10.6** A 10 GHz weather radar uses a 15-cm diameter lossless antenna. At a distance of 1 km, what are the dimensions of the volume resolvable by the radar if the pulse length is 1 μs?
- ***10.7** A radar system is characterized by the following parameters: $P_{\text{t}} = 1 \text{ kW}$, $\tau = 0.1 \text{ }\mu\text{s}$, $G = 30 \text{ dB}$, $\lambda = 3 \text{ cm}$, and $T_{\text{sys}} = 1,500 \text{ K}$. The radar cross section of a car is typically 5 m². How far away can the car be and remain detectable by the radar with a minimum signal-to-noise ratio of 13 dB?
- 10.8** A 3-cm wavelength radar is located at the origin of an x – y coordinate system. A car located at $x = 100 \text{ m}$ and $y = 200 \text{ m}$ is heading east (x direction) at a speed of 120 km/hr. What is the Doppler frequency measured by the radar?

数学与物理模型

卫星通信系统

地球静止轨道半径

$$R_0 = \left(\frac{GM_{\text{e}}T^2}{4\pi^2} \right)^{1/3}$$

接收功率

$$P_{\text{ri}} = \Upsilon(\theta) \, P_{\text{r}} = \Upsilon(\theta) \, P_{\text{t}}G_{\text{t}}G_{\text{r}} \left(\frac{\lambda}{4\pi R} \right)^2$$

噪声功率

$$P_{\text{ni}} = KT_{\text{sys}}B$$

信噪比

$$S_{\text{n}} = \frac{P_{\text{ri}}}{P_{\text{ni}}} = \frac{\Upsilon(\theta) \, P_{\text{t}}G_{\text{t}}G_{\text{r}}}{KT_{\text{sys}}B} \left(\frac{\lambda}{4\pi R} \right)^2$$

雷达传感器

最大不模糊距离

$$R_{\text{u}} = \frac{cT_{\text{p}}}{2} = \frac{c}{2f_{\text{p}}}$$

距离分辨率

$$\Delta R = R_2 - R_1 = c\tau/2$$

方位分辨率

$$\Delta x = \beta R$$

雷达方程

$$P_{\text{r}} = \frac{P_{\text{t}}G^2\lambda^2\sigma_{\text{t}}}{(4\pi)^3R^4}$$

多普勒频移

$$f_{\text{d}} = -2\frac{u_{\text{r}}}{\lambda_{\text{t}}} = -\frac{2u}{\lambda_{\text{t}}} \cos \theta$$

问题

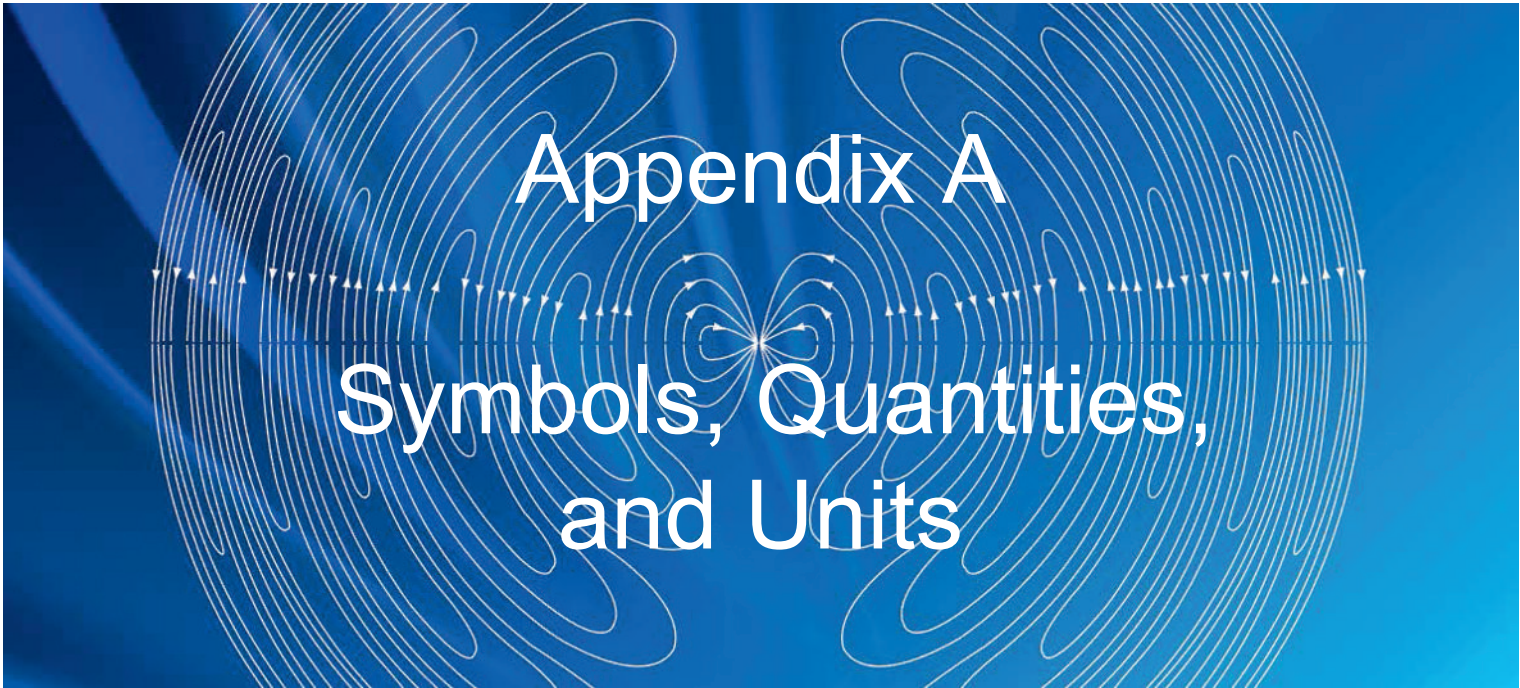
第10-1至10-4节 卫星通信系统

- *10.1 一颗遥感卫星在距地面1100 km的圆轨道上绕地球运行。求其轨道周期。
- 10.2 带宽为400 MHz的转发器采用极化分集。若每个电话信道占用4 kHz带宽，该转发器能承载多少个电话信道？
- *10.3 若每个电视信道需要6 MHz带宽，重做习题10.2。
- 10.4 一颗地球静止卫星距地面接收站40000 km。卫星发射天线为直径1 m的圆形口径，地面站抛物面天线的有效直径为20 cm。卫星在12 GHz频率上发射1 kW功率，接收机系统噪声温度为1000 K。若接收的电视信号带宽为6 MHz，求信噪比。假设天线和大气均无损。

第10-5至10-8节 雷达传感器

- *10.5 一部汽车防撞雷达需要探测最远0.5 km处的车辆。求可用的最大脉冲重复频率（PRF）。
- 10.6 一部10 GHz气象雷达采用直径15 cm的无损天线。若脉冲宽度为1 μs，求距雷达1 km处可分辨体积的各方向尺寸。
- *10.7 某雷达的参数为 $P_{\text{t}}=1 \text{ kW}$ 、 $\tau=0.1 \text{ }\mu\text{s}$ 、 $G=30 \text{ dB}$ 、 $\lambda=3 \text{ cm}$ 、 $T_{\text{sys}}=1500 \text{ K}$ 。汽车的典型雷达散射截面为5 m²。若探测所需的最小信噪比为13 dB，求仍能探测到汽车的最大距离。
- 10.8 一部波长为3 cm的雷达位于 x - y 坐标系原点。一辆汽车位于 $x=100 \text{ m}$ 、 $y=200 \text{ m}$ 处，正以120 km/h的速度向东（+ x 方向）行驶。求雷达测得的多普勒频移。

* 标记的习题在附录 E 提供答案。



Symbol	Quantity	SI Unit	Abbreviation
A	Magnetic potential (vector)	webers/meter	Wb/m
B	Susceptance	siemens	S
B	Magnetic flux density	teslas or webers/meter ²	T or W/m ²
C	Capacitance	farads	F
D	Directivity (antenna)	(dimensionless)	—
D	Electric flux density	coulombs/meter ²	C/m ²
d	Moment arm	meters	m
E	Electric-field intensity	volts/meter	V/m
<i>E</i> _{ds}	Dielectric strength	volts/meter	V/m
F	Radiation intensity (normalized)	(dimensionless)	—
F	Force	newtons	N
<i>f</i>	Frequency	hertz	Hz
<i>f</i> _d	Doppler frequency	hertz	Hz
<i>f</i> _{mn}	Cutoff frequency	hertz	Hz
G	Conductance	siemens	S
G	Gain (power)	(dimensionless)	—
H	Magnetic field intensity	amperes/meter	A/m
I	Current	amperes	A
J	Current density (volume)	amperes/meter ²	A/m ²
J _s	Current density (surface)	amperes/meter	A/m
k	Wavenumber	radians/meter	rad/m
<i>k</i> _c	Cutoff wavenumber	radians/second	rad/s
L	Inductance	henrys	H
l	Length	meters	m

附录 A 符号、物理量与单位

粗体表示矢量；同一字母可在不同上下文中代表不同物理量。

符号	物理量	SI 单位	单位符号
A	磁矢势	韦伯/米	Wb/m
B	电纳	西门子	S
B	磁通密度	特斯拉；韦伯/平方米	T； Wb/m ²
C	电容	法拉	F
D	天线方向性系数	无量纲	-
D	电通量密度	库仑/平方米	C/m ²
d	力臂	米	m
E	电场强度	伏特/米	V/m
<i>E</i> _{ds}	介电强度	伏特/米	V/m
F	归一化辐射强度	无量纲	-
F	力	牛顿	N
f	频率	赫兹	Hz
<i>f</i> _d	多普勒频率	赫兹	Hz
<i>f</i> _{mn}	截止频率	赫兹	Hz
G	电导	西门子	S
G	功率增益	无量纲	-
H	磁场强度	安培/米	A/m
I	电流	安培	A
J	体电流密度	安培/平方米	A/m ²
J _s	面电流密度	安培/米	A/m
k	波数	弧度/米	rad/m
<i>k</i> _c	截止波数	弧度/米	rad/m
L	电感	亨利	H
l	长度	米	m

校注：磁通密度单位为 T = Wb/m²，原书缩写栏漏印 b；截止波数 *k*_c 的单位为 rad/m，原书误印为 rad/s。两处按物理量定义更正。

Symbol	Quantity	SI Unit	Abbreviation
M, m	Mass	kilograms	kg
$\mathbf{M}$	Magnetization vector	amperes/meter	A/m
$\mathbf{m}$	Magnetic dipole moment	ampere-meters ²	A·m ²
n	Index of refraction	(dimensionless)	—
P	Power	watts	W
$\mathbf{P}$	Electric polarization vector	coulombs/meter ²	C/m ²
p	Pressure	newtons/meter ²	N/m ²
$\mathbf{p}$	Electric dipole moment	coulomb-meters	C·m
Q	Quality factor	(dimensionless)	—
Q, q	Charge	coulombs	C
R	Reflectivity (reflectance)	(dimensionless)	—
R	Resistance	ohms	Ω
R	Range	meters	m
r	Radial distance	meters	m
S	Standing-wave ratio	(dimensionless)	—
$\mathbf{S}$	Poynting vector	watts/meter ²	W/m ²
S_{av}	Power density	watts/meter ²	W/m ²
T	Temperature	kelvins	K
T	Transmissivity (transmittance)	(dimensionless)	—
$\mathbf{T}$	Torque	newton-meters	N·m
t	Time	seconds	s
T	period	seconds	s
$\mathbf{u}$	Velocity	meters/second	m/s
u_{g}	Group velocity	meters/second	m/s
u_{p}	Phase velocity	meters/second	m/s
V	Electric potential	volts	V
V	Voltage	volts	V
V_{bv}	Voltage breakdown	volts	V
V_{emf}	Electromotive force (emf)	volts	V
W	Energy (work)	joules	J
w	Energy density	joules/meter ³	J/m ³
X	Reactance	ohms	Ω
Y	Admittance	siemens	S
Z	Impedance	ohms	Ω
α	Attenuation constant	nepers/meter	Np/m
β	Beamwidth	degrees	°
β	Phase constant (wavenumber)	radians/meter	rad/m
Γ	Reflection coefficient	(dimensionless)	—
γ	Propagation constant	meters ^{−1}	m ^{−1}
δ_{s}	Skin depth	meters	m
ϵ, ϵ_0	Permittivity	farads/meter	F/m
ϵ_{r}	Relative permittivity	(dimensionless)	—
η	Impedance	ohms	Ω
λ	Wavelength	meters	m

附录 A 符号、物理量与单位

粗体表示矢量；同一字母可在不同上下文中代表不同物理量。

符号	物理量	SI 单位	单位符号
M, m	质量	千克	kg
M	磁化强度矢量	安培/米	A/m
m	磁偶极矩	安培·平方米	A·m ²
n	折射率	无量纲	-
P	功率	瓦特	W
P	电极化强度矢量	库仑/平方米	C/m ²
p	压强	牛顿/平方米	N/m ²
p	电偶极矩	库仑·米	C·m
Q	品质因数	无量纲	-
Q, q	电荷量	库仑	C
R	反射率	无量纲	-
R	电阻	欧姆	Ω
R	距离	米	m
r	径向距离	米	m
S	驻波比	无量纲	-
S	坡印廷矢量	瓦特/平方米	W/m ²
S _{av}	平均功率密度	瓦特/平方米	W/m ²
T	温度	开尔文	K
T	透射率	无量纲	-
T	力矩	牛顿·米	N·m
t	时间	秒	s
T	周期	秒	s
u	速度	米/秒	m/s
u _g	群速度	米/秒	m/s
u _p	相速度	米/秒	m/s
V	电势	伏特	V
V	电压	伏特	V
V _{bv}	击穿电压	伏特	V
V _{emf}	电动势	伏特	V
W	能量（功）	焦耳	J
w	能量密度	焦耳/立方米	J/m ³
X	电抗	欧姆	Ω
Y	导纳	西门子	S
Z	阻抗	欧姆	Ω
α	衰减常数	奈培/米	Np/m
β	波束宽度	度	°
β	相位常数（波数）	弧度/米	rad/m
Γ	反射系数	无量纲	-
γ	传播常数	每米	m ^{−1}
δ _s	趋肤深度	米	m
ε, ε ₀	介电率	法拉/米	F/m
ε _r	相对介电率	无量纲	-
η	波阻抗	欧姆	Ω
λ	波长	米	m

Symbol	Quantity	SI Unit	Abbreviation
μ, μ_0	Permeability	henrys/meter	H/m
μ_r	Relative permeability	(dimensionless)	—
μ_e, μ_h	Mobility (electron, hole)	meters ² /volt·second	m ² /V·s
ρ_l	Charge density (linear)	coulombs/meter	C/m
ρ_s	Charge density (surface)	coulombs/meter ²	C/m ²
ρ_v	Charge density (volume)	coulombs/meter ³	C/m ³
σ	Conductivity	siemens/meter	S/m
σ_t	Radar cross section	meters ²	m ²
τ	Transmission coefficient	(dimensionless)	—
τ	Pulse length	seconds	s
Υ	Atmospheric transmissivity	(dimensionless)	—
Φ	Magnetic flux	webers	Wb
Ψ	Gravitational field	newtons/kilogram	N/kg
χ_e	Electric susceptibility	(dimensionless)	—
χ_m	Magnetic susceptibility	(dimensionless)	—
Ω	Solid angle	steradians	sr
ω	Angular frequency	radians/second	rad/s
ω	Angular velocity	radians/second	rad/s

附录 A 符号、物理量与单位

粗体表示矢量；同一字母可在不同上下文中代表不同物理量。

符号	物理量	SI 单位	单位符号
μ, μ_0	磁导率	亨利/米	H/m
μ_r	相对磁导率	无量纲	-
μ_e, μ_h	迁移率 (电子、空穴)	平方米/(伏特·秒)	m ² /(V·s)
ρ_l	线电荷密度	库仑/米	C/m
ρ_s	面电荷密度	库仑/平方米	C/m ²
ρ_v	体电荷密度	库仑/立方米	C/m ³
σ	电导率	西门子/米	S/m
σ_t	雷达散射截面	平方米	m ²
τ	透射系数	无量纲	-
τ	脉冲宽度	秒	s
Υ	大气透射率	无量纲	-
Φ	磁通量	韦伯	Wb
Ψ	引力场强度	牛顿/千克	N/kg
χ_e	电极化率	无量纲	-
χ_m	磁化率	无量纲	-
Ω	立体角	球面度	sr
ω	角频率	弧度/秒	rad/s
ω	角速度	弧度/秒	rad/s



TABLE B-1
RELATIVE PERMITTIVITY ϵ_r
OF COMMON MATERIALS^a

$$\epsilon = \epsilon_r \epsilon_0 \text{ and } \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m.}$$

Material	Relative Permittivity, ϵ_r	Material	Relative Permittivity, ϵ_r
Vacuum	1	Dry soil	2.5–3.5
Air (at sea level)	1.0006	Plexiglass	3.4
Styrofoam	1.03	Glass	4.5–10
Teflon	2.1	Quartz	3.8–5
Petroleum oil	2.1	Bakelite	5
Wood (dry)	1.5–4	Porcelain	5.7
Paraffin	2.2	Formica	6
Polyethylene	2.25	Mica	5.4–6
Polystyrene	2.6	Ammonia	22
Paper	2–4	Seawater	72–80
Rubber	2.2–4.1	Distilled water	81

^aThese are low-frequency values at room temperature (20 °C).

Note: For most metals, $\epsilon_r \approx 1$.

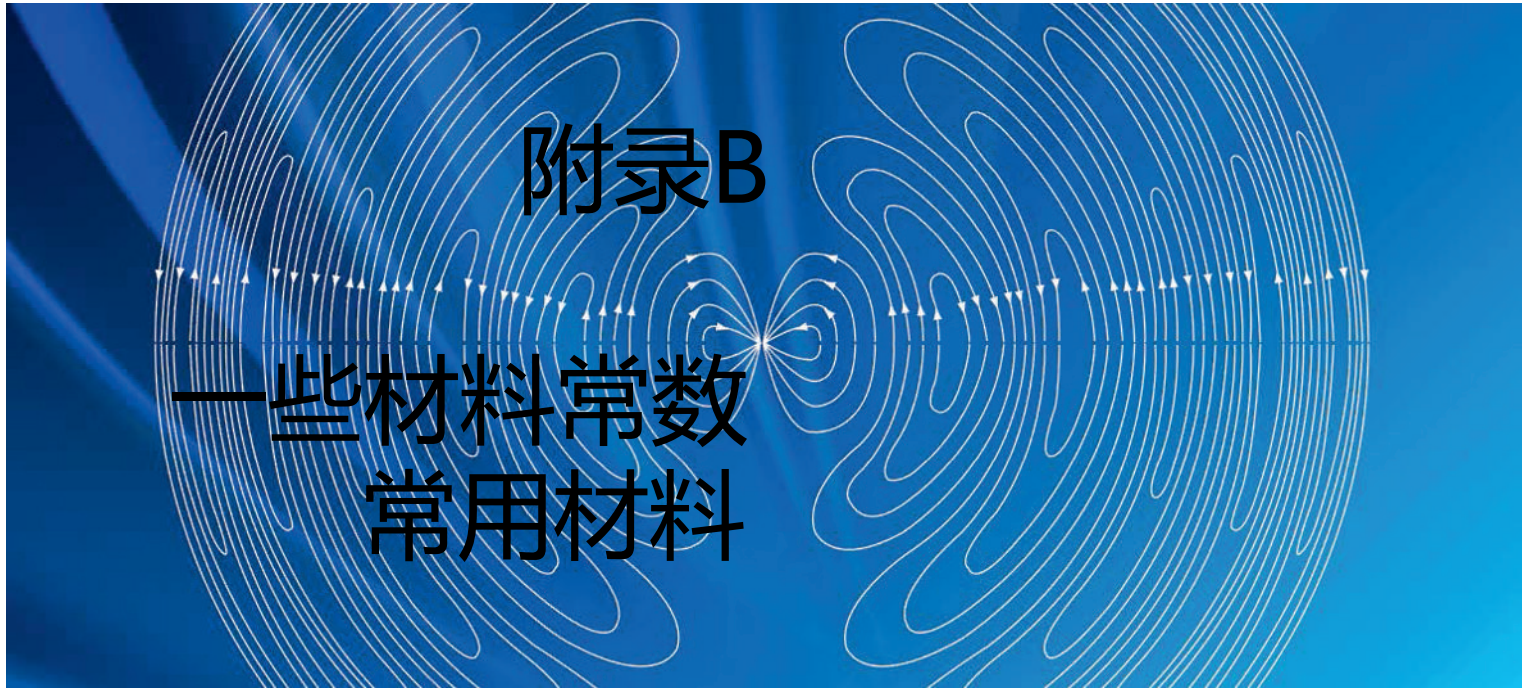


表 B-1 常见材料^a 的相对介电常数 ϵ_r

$$\epsilon = \epsilon_r \epsilon_0 \text{ 和 } \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m.}$$

材料	相对的介电常数, ϵ_r	材料	相对的介电常数, ϵ_r
真空	1	干燥的土壤	2.5–3.5
空气 (海平面)	1.0006	有机玻璃	3.4
聚苯乙烯泡沫塑料	1.03	玻璃	4.5–10
Teflon	2.1	石英	3.8–5
石油类	2.1	胶木	5
木材 (干)	1.5–4	瓷	5.7
石蜡	2.2	福米卡	6
聚乙烯	2.25	云母	5.4–6
聚苯乙烯	2.6	氨	22
纸	2–4	海水	72–80
橡皮	2.2–4.1	蒸馏水	81

^a这些是室温下的低频值 (20 °C).

注意: 对于大多数金属, $\epsilon_r \approx 1$ 。

TABLE B-2
 CONDUCTIVITY σ OF SOME COMMON MATERIALS^a

Material	Conductivity σ (S/m)	Material	Conductivity σ (S/m)
Conductors		Semiconductors	
Silver	6.2×10^7	Pure germanium	2.2
Copper	5.8×10^7	Pure silicon	4.4×10^{-4}
Gold	4.1×10^7	Insulators	
Aluminum	3.5×10^7	Wet soil	$\sim 10^{-2}$
Tungsten	1.8×10^7	Fresh water	$\sim 10^{-3}$
Zinc	1.7×10^7	Distilled water	$\sim 10^{-4}$
Brass	1.5×10^7	Dry soil	$\sim 10^{-4}$
Iron	10^7	Glass	10^{-12}
Bronze	10^7	Hard rubber	10^{-15}
Tin	9×10^6	Paraffin	10^{-15}
Lead	5×10^6	Mica	10^{-15}
Mercury	10^6	Fused quartz	10^{-17}
Carbon	3×10^4	Wax	10^{-17}
Seawater	4		
Animal body (average)	0.3 (poor cond.)		
^a These are low-frequency values at room temperature (20 °C).			

常用材料的电导率

表 B-2 常用材料的电导率 σ

材料	电导率 σ (S/m)	材料	电导率 σ (S/m)
导体		半导体	
银	6.2×10^7	纯锗	2.2
铜	5.8×10^7	纯硅	4.4×10^{-4}
金	4.1×10^7	绝缘体	
铝	3.5×10^7	湿土	约 10^{-2}
钨	1.8×10^7	淡水	约 10^{-3}
锌	1.7×10^7	蒸馏水	约 10^{-4}
黄铜	1.5×10^7	干土	约 10^{-4}
铁	10^7	玻璃	10^{-12}
青铜	10^7	硬橡胶	10^{-15}
锡	9×10^6	石蜡	10^{-15}
铅	5×10^6	云母	10^{-15}
汞	10^6	熔融石英	10^{-17}
碳	3×10^4	蜡	10^{-17}
海水	4		
动物体（平均）	0.3（不良导体）		

注：表中数据为室温（20 °C）下的低频值。约号表示近似值。

TABLE B-3
RELATIVE PERMEABILITY
 μ_r OF SOME COMMON MATERIALS^a

$\mu = \mu_r \mu_0$ and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

Material	Relative Permeability, μ_r
Diamagnetic	
Bismuth	$0.99983 \approx 1$
Gold	$0.99996 \approx 1$
Mercury	$0.99997 \approx 1$
Silver	$0.99998 \approx 1$
Copper	$0.99999 \approx 1$
Water	$0.99999 \approx 1$
Paramagnetic	
Air	$1.000004 \approx 1$
Aluminum	$1.00002 \approx 1$
Tungsten	$1.00008 \approx 1$
Titanium	$1.0002 \approx 1$
Platinum	$1.0003 \approx 1$
Ferromagnetic (nonlinear)	
Cobalt	250
Nickel	600
Mild steel	2,000
Iron (pure)	4,000–5,000
Silicon iron	7,000
Mumetal	$\sim 100,000$
Purified iron	$\sim 200,000$
^a These are typical values; actual values depend on material variety.	

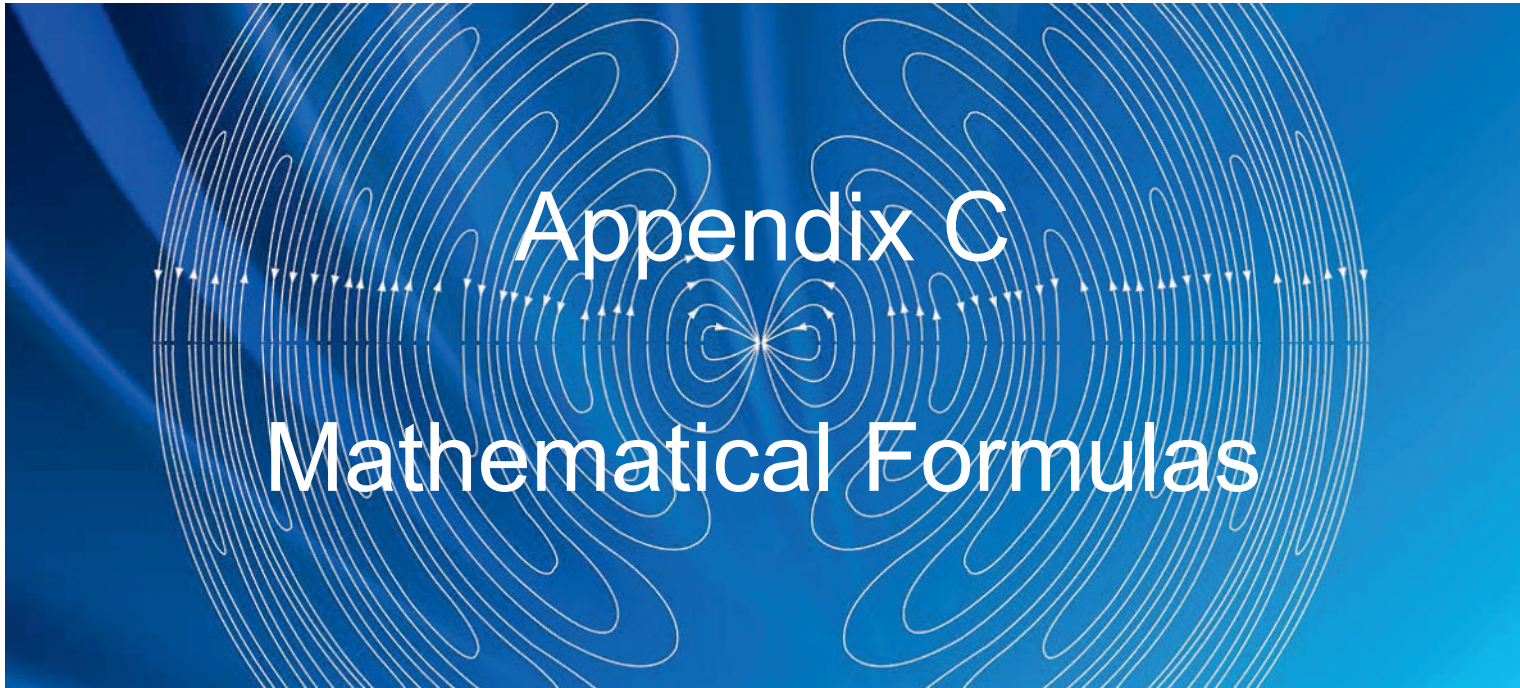
Note: Except for ferromagnetic materials, $\mu_r \approx 1$ for all dielectrics and conductors.

表 B-3 一些常见材料的相对磁导率 μ_r ^a

$\mu = \mu_r \mu_0$ 和 $\mu_0 = 4\pi \times 10^{-7}$ H/m。

材料	相对的渗透性, μ_r
抗磁性	
铋	$0.99983 \approx 1$
金	$0.99996 \approx 1$
汞	$0.99997 \approx 1$
银	$0.99998 \approx 1$
铜	$0.99999 \approx 1$
水	$0.99999 \approx 1$
顺磁性	
Air	$1.000004 \approx 1$
铝	$1.00002 \approx 1$
钨	$1.00008 \approx 1$
钛	$1.0002 \approx 1$
铂	$1.0003 \approx 1$
铁磁 (非线性)	
钴	250
镍	600
低碳钢	2,000
铁 (纯)	4,000–5,000
硅铁	7,000
镍铁合金	$\sim 100,000$
精铁	$\sim 200,000$
^a 这些是典型值；实际值取决于材料的品种。	

注：除铁磁材料外， $\mu_r \approx 1$ 适用于所有电介质和导体。



Trigonometric Relations

$$\begin{aligned} \sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y \\ \cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y \\ 2 \sin x \sin y &= \cos(x - y) - \cos(x + y) \\ 2 \sin x \cos y &= \sin(x + y) + \sin(x - y) \\ 2 \cos x \cos y &= \cos(x + y) + \cos(x - y) \\ \sin 2x &= 2 \sin x \cos x \\ \cos 2x &= 1 - 2 \sin^2 x \\ \sin x + \sin y &= 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ \sin x - \sin y &= 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \\ \cos x + \cos y &= 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ \cos x - \cos y &= -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \\ \cos(x \pm 90^\circ) &= \mp \sin x \\ \cos(-x) &= \cos x \\ \sin(x \pm 90^\circ) &= \pm \cos x \\ \sin(-x) &= -\sin x \\ e^{jx} &= \cos x + j \sin x \quad (\text{Euler's identity}) \\ \sin x &= \frac{e^{jx} - e^{-jx}}{2j} \\ \cos x &= \frac{e^{jx} + e^{-jx}}{2} \end{aligned}$$

Approximations for Small Quantities

$$\begin{aligned} \text{For } |x| \ll 1, \\ (1 \pm x)^n &\approx 1 \pm nx \\ (1 \pm x)^2 &\approx 1 \pm 2x \\ \sqrt{1 \pm x} &\approx 1 \pm \frac{x}{2} \\ \frac{1}{\sqrt{1 \pm x}} &\approx 1 \mp \frac{x}{2} \\ e^x &= 1 + x + \frac{x^2}{2!} + \cdots \approx 1 + x \\ \ln(1+x) &\approx x \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \approx x \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \approx 1 - \frac{x^2}{2} \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1 \end{aligned}$$



三角恒等式

$$\begin{aligned} \sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y \\ \cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y \\ 2 \sin x \sin y &= \cos(x - y) - \cos(x + y) \\ 2 \sin x \cos y &= \sin(x + y) + \sin(x - y) \\ 2 \cos x \cos y &= \cos(x + y) + \cos(x - y) \\ \sin 2x &= 2 \sin x \cos x \\ \cos 2x &= 1 - 2 \sin^2 x \\ \sin x + \sin y &= 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ \sin x - \sin y &= 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \\ \cos x + \cos y &= 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ \cos x - \cos y &= -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \\ \cos(x \pm 90^\circ) &= \mp \sin x \\ \cos(-x) &= \cos x \\ \sin(x \pm 90^\circ) &= \pm \cos x \\ \sin(-x) &= -\sin x \\ e^{jx} &= \cos x + j \sin x \quad (\text{欧拉恒等式}) \\ \sin x &= \frac{e^{jx} - e^{-jx}}{2j} \\ \cos x &= \frac{e^{jx} + e^{-jx}}{2} \end{aligned}$$

小量近似与级数展开

$$\begin{aligned} \text{当 } |x| \text{ 远小于 } 1 \text{ 时:} \\ (1 \pm x)^n &\approx 1 \pm nx \\ (1 \pm x)^2 &\approx 1 \pm 2x \\ \sqrt{1 \pm x} &\approx 1 \pm \frac{x}{2} \\ \frac{1}{\sqrt{1 \pm x}} &\approx 1 \mp \frac{x}{2} \\ e^x &= 1 + x + \frac{x^2}{2!} + \cdots \approx 1 + x \\ \ln(1+x) &\approx x \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \approx x \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \approx 1 - \frac{x^2}{2} \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1 \end{aligned}$$

Common Derivatives

$f(x)$	$\frac{df}{dx}$
x^a	ax^{a-1}
$\frac{1}{x^a}$	$-ax^{-(a+1)}$
$\sqrt{x}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
e^{ax}	ae^{ax}
b^x	$b^x \ln(b)$
$\ln(x)$	$\frac{1}{x}$
$\log_a(x)$	$\frac{1}{x \ln(a)}$
$\sin(ax)$	$a \cos(ax)$
$\cos(ax)$	$-a \sin(ax)$
$\tan(ax)$	$a \sec^2(ax)$

Common Integrals

$\int x^n \, dx$	$= \frac{1}{n+1} x^{n+1} + C$
$\int \frac{1}{x} \, dx$	$= \ln x + C$
$\int x^{-n} \, dx$	$= \frac{1}{-n+1} x^{-n+1} + C$
$\int \frac{1}{ax+b} \, dx$	$= \frac{1}{a} \ln ax+b + C$
$\int \cos(ax) \, dx$	$= \frac{1}{a} \sin(ax) + C$
$\int \sin(ax) \, dx$	$= -\frac{1}{a} \cos(ax) + C$
$\int \tan(ax) \, dx$	$= \frac{1}{a} \ln \sec(ax) + C$
$\int e^{ax} \, dx$	$= \frac{1}{a} e^{ax} + C$
$\int b^x \, dx$	$= \frac{b^x}{\ln(b)} + C$
$\int \ln x \, dx$	$= x \ln(x) - x + C$
$\int e^{ax} \sin(bx) \, dx$	$= \frac{e^{ax}}{a^2+b^2} (a \sin(bx) - b \cos(bx)) + C$
$\int e^{ax} \cos(bx) \, dx$	$= \frac{e^{ax}}{a^2+b^2} (a \cos(bx) + b \sin(bx)) + C$

$$\int x e^x \, dx = (x-1)e^x + C$$

Gradient, Divergence, Curl, and Laplacian Operators

Cartesian (Rectangular) Coordinates (x , y , z)

$$\begin{aligned} \nabla V &= \hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \\ \nabla \cdot \mathbf{A} &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \\ \nabla \times \mathbf{A} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \\ &\quad + \hat{\mathbf{y}} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &\quad + \hat{\mathbf{z}} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\ \nabla^2 V &= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \end{aligned}$$

Cylindrical Coordinates (r , ϕ , z)

$$\begin{aligned} \nabla V &= \hat{\mathbf{r}} \frac{\partial V}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \\ \nabla \cdot \mathbf{A} &= \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \\ \nabla \times \mathbf{A} &= \frac{1}{r} \begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix} \\ &= \hat{\mathbf{r}} \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \\ &\quad + \hat{\boldsymbol{\phi}} \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \\ &\quad + \hat{\mathbf{z}} \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right] \\ \nabla^2 V &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} \end{aligned}$$

常用导数

$f(x)$	$\frac{df}{dx}$
x^a	ax^{a-1}
$\frac{1}{x^a}$	$-ax^{-(a+1)}$
$\sqrt{x}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
e^{ax}	ae^{ax}
b^x	$b^x \ln(b)$
$\ln(x)$	$\frac{1}{x}$
$\log_a(x)$	$\frac{1}{x \ln(a)}$
$\sin(ax)$	$a \cos(ax)$
$\cos(ax)$	$-a \sin(ax)$
$\tan(ax)$	$a \sec^2(ax)$

常用积分

$\int x^n \, dx$	$= \frac{1}{n+1} x^{n+1} + C$
$\int \frac{1}{x} \, dx$	$= \ln x + C$
$\int x^{-n} \, dx$	$= \frac{1}{-n+1} x^{-n+1} + C$
$\int \frac{1}{ax+b} \, dx$	$= \frac{1}{a} \ln ax+b + C$
$\int \cos(ax) \, dx$	$= \frac{1}{a} \sin(ax) + C$
$\int \sin(ax) \, dx$	$= -\frac{1}{a} \cos(ax) + C$
$\int \tan(ax) \, dx$	$= \frac{1}{a} \ln \sec(ax) + C$
$\int e^{ax} \, dx$	$= \frac{1}{a} e^{ax} + C$
$\int b^x \, dx$	$= \frac{b^x}{\ln(b)} + C$
$\int \ln x \, dx$	$= x \ln(x) - x + C$
$\int e^{ax} \sin(bx) \, dx$	$= \frac{e^{ax}}{a^2+b^2} (a \sin(bx) - b \cos(bx)) + C$
$\int e^{ax} \cos(bx) \, dx$	$= \frac{e^{ax}}{a^2+b^2} (a \cos(bx) + b \sin(bx)) + C$

$$\int x e^x \, dx = (x-1)e^x + C$$

梯度、散度、旋度与拉普拉斯算子

直角坐标 (x , y , z)

$$\begin{aligned} \nabla V &= \hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \\ \nabla \cdot \mathbf{A} &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \\ \nabla \times \mathbf{A} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \\ &\quad + \hat{\mathbf{y}} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &\quad + \hat{\mathbf{z}} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\ \nabla^2 V &= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \end{aligned}$$

圆柱坐标 (r , ϕ , z)

$$\begin{aligned} \nabla V &= \hat{\mathbf{r}} \frac{\partial V}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial V}{\partial z} \\ \nabla \cdot \mathbf{A} &= \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \\ \nabla \times \mathbf{A} &= \frac{1}{r} \begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix} \\ &= \hat{\mathbf{r}} \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \\ &\quad + \hat{\boldsymbol{\phi}} \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \\ &\quad + \hat{\mathbf{z}} \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right] \\ \nabla^2 V &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} \end{aligned}$$

Spherical Coordinates (R, θ, ϕ)

$$\begin{aligned} \nabla V &= \hat{\mathbf{R}} \frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi} \\ \nabla \cdot \mathbf{A} &= \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi} \\ \nabla \times \mathbf{A} &= \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} R & \hat{\boldsymbol{\phi}} R \sin \theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix} \\ &= \hat{\mathbf{R}} \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \\ &\quad + \hat{\boldsymbol{\theta}} \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right] \\ &\quad + \hat{\boldsymbol{\phi}} \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \\ \nabla^2 V &= \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) \\ &\quad + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) \\ &\quad + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} \end{aligned}$$

Some Useful Vector Identities

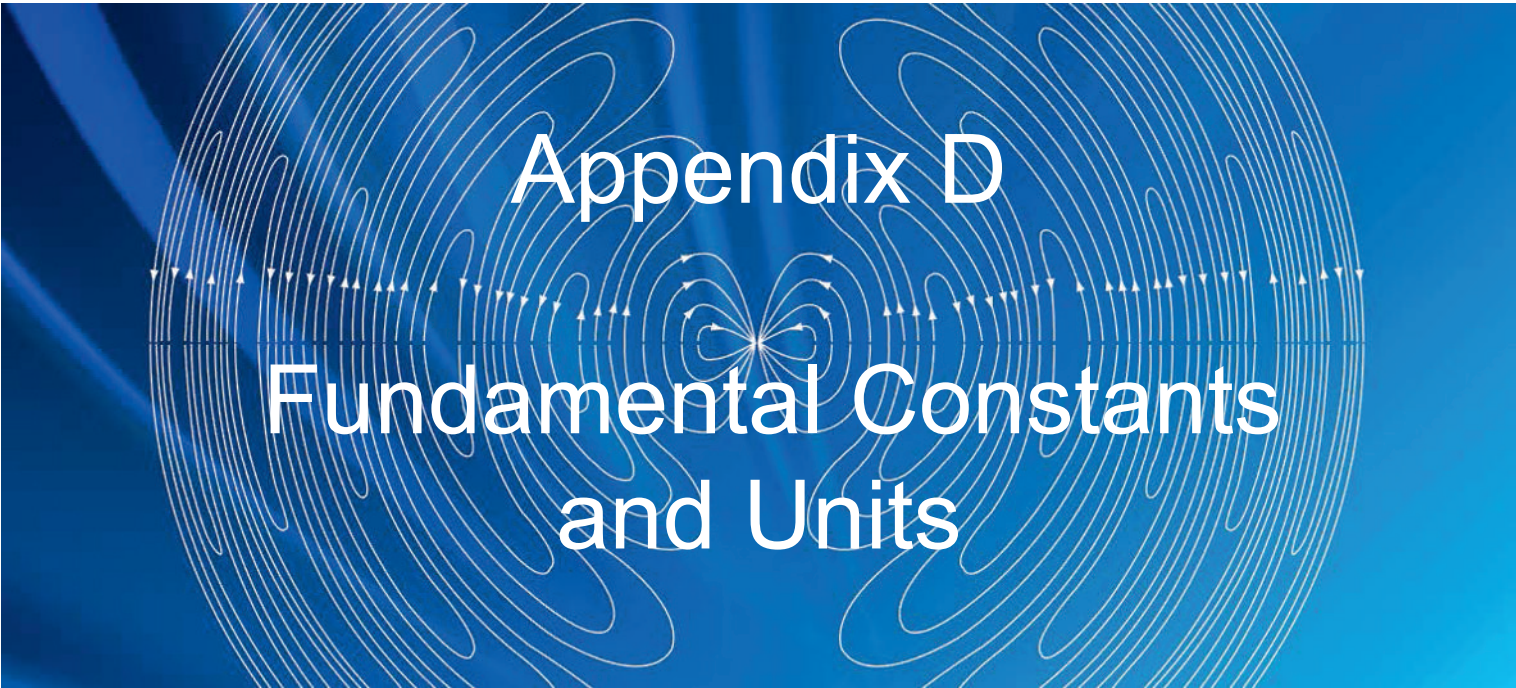
$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= AB \cos \theta_{AB} && \text{Scalar (or dot) product} \\ \mathbf{A} \times \mathbf{B} &= \hat{\mathbf{n}} AB \sin \theta_{AB} && \text{Vector (or cross) product, } \hat{\mathbf{n}} \text{ normal to plane containing } \mathbf{A} \text{ and } \mathbf{B} \\ \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \\ \nabla(U + V) &= \nabla U + \nabla V \\ \nabla(UV) &= U \nabla V + V \nabla U \\ \nabla \cdot (\mathbf{A} + \mathbf{B}) &= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} \\ \nabla \cdot (U \mathbf{A}) &= U \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla U \\ \nabla \times (U \mathbf{A}) &= U \nabla \times \mathbf{A} + \nabla U \times \mathbf{A} \\ \nabla \times (\mathbf{A} + \mathbf{B}) &= \nabla \times \mathbf{A} + \nabla \times \mathbf{B} \\ \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\ \nabla \cdot (\nabla \times \mathbf{A}) &= 0 \\ \nabla \times \nabla V &= 0 \\ \nabla \cdot \nabla V &= \nabla^2 V \\ \nabla \times \nabla \times \mathbf{A} &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \\ \int_v (\nabla \cdot \mathbf{A}) d\mathcal{V} &= \oint_S \mathbf{A} \cdot d\mathbf{s} && \text{Divergence theorem} \\ &&& (S \text{ encloses } \mathcal{V}) \\ \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} &= \oint_C \mathbf{A} \cdot d\mathbf{l} && \text{Stokes's theorem} \\ &&& (S \text{ bounded by } C) \end{aligned}$$

球坐标 (R, θ, φ)

$$\begin{aligned} \nabla V &= \hat{\mathbf{R}} \frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi} \\ \nabla \cdot \mathbf{A} &= \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi} \\ \nabla \times \mathbf{A} &= \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} R & \hat{\boldsymbol{\phi}} R \sin \theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix} \\ &= \hat{\mathbf{R}} \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \\ &\quad + \hat{\boldsymbol{\theta}} \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right] \\ &\quad + \hat{\boldsymbol{\phi}} \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \\ \nabla^2 V &= \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) \\ &\quad + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) \\ &\quad + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} \end{aligned}$$

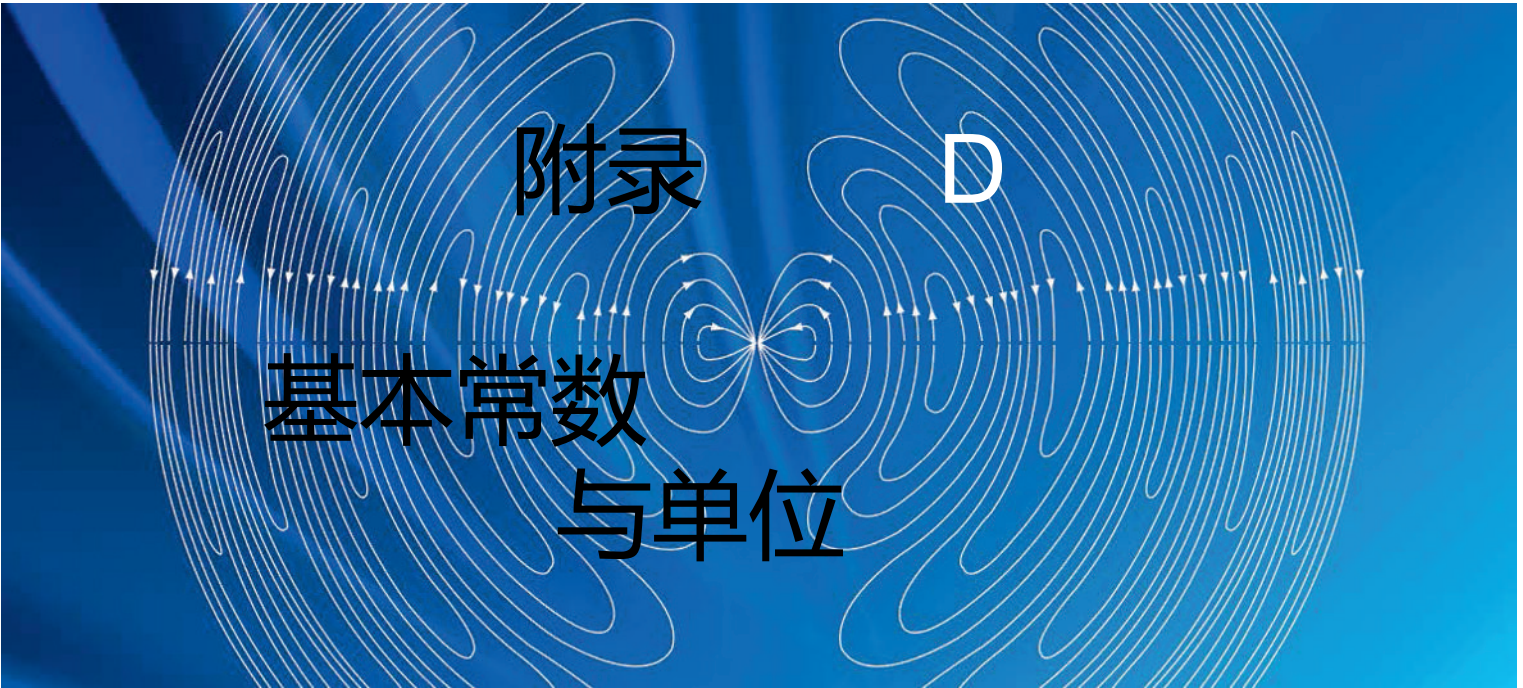
常用矢量恒等式

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= AB \cos \theta_{AB} && \text{标量积 (点积)} \\ \mathbf{A} \times \mathbf{B} &= \hat{\mathbf{n}} AB \sin \theta_{AB} && \text{矢量积 (叉积)} \\ &&& \text{单位法向量垂直于 } \mathbf{A}、\mathbf{B} \text{ 所在的平面。} \\ \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \\ \nabla(U + V) &= \nabla U + \nabla V \\ \nabla(UV) &= U \nabla V + V \nabla U \\ \nabla \cdot (\mathbf{A} + \mathbf{B}) &= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} \\ \nabla \cdot (U \mathbf{A}) &= U \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla U \\ \nabla \times (U \mathbf{A}) &= U \nabla \times \mathbf{A} + \nabla U \times \mathbf{A} \\ \nabla \times (\mathbf{A} + \mathbf{B}) &= \nabla \times \mathbf{A} + \nabla \times \mathbf{B} \\ \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\ \nabla \cdot (\nabla \times \mathbf{A}) &= 0 \\ \nabla \times \nabla V &= 0 \\ \nabla \cdot \nabla V &= \nabla^2 V \\ \nabla \times \nabla \times \mathbf{A} &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \\ \int_v (\nabla \cdot \mathbf{A}) d\mathcal{V} &= \oint_S \mathbf{A} \cdot d\mathbf{s} && \text{散度定理} \\ &&& (S \text{ 包围体积 } \mathcal{V}) \\ \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} &= \oint_C \mathbf{A} \cdot d\mathbf{l} && \text{斯托克斯定理} \\ &&& (S \text{ 以 } C \text{ 为边界}) \end{aligned}$$



FUNDAMENTAL PHYSICAL CONSTANTS

CONSTANT	SYMBOL	VALUE
Speed of light in vacuum	c	$2.998 \times 10^8 \approx 3 \times 10^8$ m/s
Gravitational constant	G	6.67×10^{-11} N·m ² /kg ²
Boltzmann’s constant	K	1.38×10^{-23} J/K
Elementary charge	e	1.60×10^{-19} C
Permittivity of free space	ϵ_0	$8.85 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9}$ F/m
Permeability of free space	μ_0	$4\pi \times 10^{-7}$ H/m
Electron mass	m_e	9.11×10^{-31} kg
Proton mass	m_p	1.67×10^{-27} kg
Planck’s constant	h	6.63×10^{-34} J·s
Intrinsic impedance of free space	η_0	$376.7 \approx 120\pi$ Ω



基本物理常数

常数	符号	数值
真空中的光速	c	$2.998 \times 10^8 \approx 3 \times 10^8$ m/s
万有引力常数	G	6.67×10^{-11} N·m ² /kg ²
玻尔兹曼常数	K	1.38×10^{-23} J/K
元电荷	e	1.60×10^{-19} C
真空介电常数	ϵ_0	$8.85 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9}$ F/m
真空磁导率	μ_0	$4\pi \times 10^{-7}$ H/m
电子质量	m_e	9.11×10^{-31} kg
质子质量	m_p	1.67×10^{-27} kg
普朗克常数	h	6.63×10^{-34} J·s
真空本征阻抗	η_0	$376.7 \approx 120\pi$ Ω

FUNDAMENTAL SI UNITS

DIMENSION	UNIT	SYMBOL
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol

MULTIPLE AND SUBMULTIPLE PREFIXES

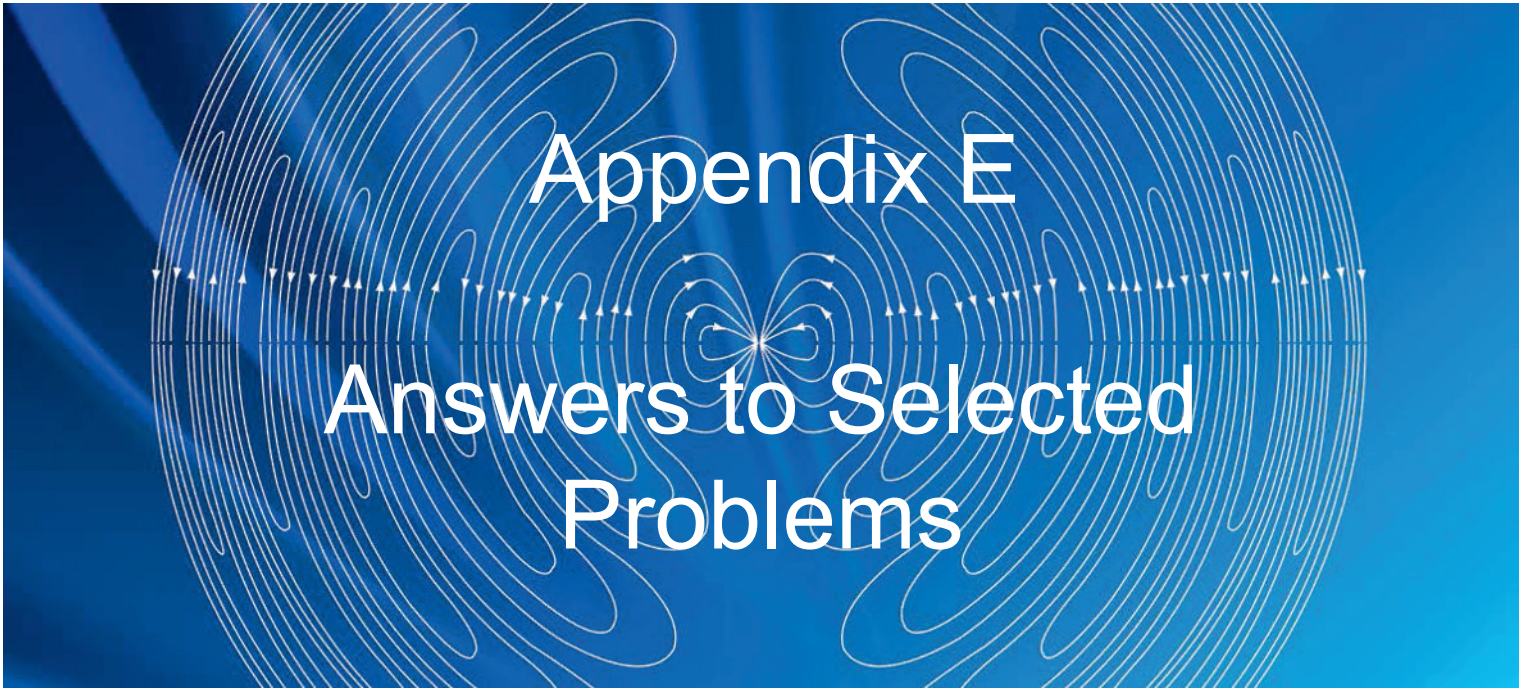
PREFIX	SYMBOL	MAGNITUDE	PREFIX	SYMBOL	MAGNITUDE
exa	E	10 ¹⁸	milli	m	10 ^{−3}
peta	P	10 ¹⁵	micro	μ	10 ^{−6}
tera	T	10 ¹²	nano	n	10 ^{−9}
giga	G	10 ⁹	pico	p	10 ^{−12}
mega	M	10 ⁶	femto	f	10 ^{−15}
kilo	k	10 ³	atto	a	10 ^{−18}

SI基本单位

量纲	单位	符号
长度	米	m
质量	千克	kg
时间	秒	s
电流	安培	A
温度	开尔文	K
物质的量	摩尔	mol

十进制倍数与分数词头

词头	符号	倍率	词头	符号	倍率
exa	E	10 ¹⁸	毫	m	10 ^{−3}
拍	P	10 ¹⁵	微	μ	10 ^{−6}
太	T	10 ¹²	纳	n	10 ^{−9}
吉	G	10 ⁹	皮	p	10 ^{−12}
兆	M	10 ⁶	飞	f	10 ^{−15}
千	k	10 ³	阿	a	10 ^{−18}



Chapter 1

- 1.1

10 cm
- 1.3

$p(x,t) = 32.36\cos(4\pi \times 10^3t - 12.12\pi x + 36^\circ)$ (N/m²)
- 1.6

$u_p = 0.83$ (m/s); $\lambda = 10.47$ m
- 1.8

(a) $y_1(x,t)$ is traveling in positive x direction. $y_2(x,t)$ is traveling in negative x direction.
- 1.10

$\lambda = 1.4$ m
- 1.12

$y_2(t)$ lags $y_1(t)$ by 30° .
- 1.14

$\alpha = 2 \times 10^{-3}$ (Np/m)
- 1.17

(c) $z_1z_2 = 18e^{j109.4^\circ}$
- 1.18

(b) $z_2 = 1.22(-1 + j)$
- 1.20

(c) $|z|^2 = 20$
- 1.21

(d) $t = 0$; $s = 6e^{j30^\circ}$
- 1.23

$\ln(z) = 1.76 - j1.03$
- 1.26

$v_c(t) = 15.57\cos(2\pi \times 10^3t - 81.5^\circ)$ V
- 1.27

(d) $i(t) = 3.61\cos(\omega t + 146.31^\circ)$ A
- 1.28

(d) $\tilde{I} = -2e^{j3\pi/4} = 2e^{-j\pi}e^{j3\pi/4} = 2e^{-j\pi/4}$ A

Chapter 2

- 2.1

(a) $l/\lambda = 1.33 \times 10^{-5}$; transmission line may be ignored.
(c) $l/\lambda = 0.4$; transmission line effects should be included.
- 2.4

$R' = 1.38$ (Ω /m), $L' = 1.57 \times 10^{-7}$ (H/m), $G' = 0$, $C' = 1.84 \times 10^{-10}$ (F/m)
- 2.8

$\alpha = 0.109$ Np/m; $\beta = 44.5$ rad/m; $Z_0 = (19.6 + j0.030)$ Ω ; $u_p = 1.41 \times 10^8$ m/s
- 2.10

$w = 1.542$ mm, $\lambda = 0.044$ m

- 2.14

$R' = 1$ (Ω /m); $L' = 200$ (nH/m); $G' = 400$ (μ S/m); $C' = 80$ (pF/m); $\lambda = 2.5$ m
- 2.16

$R' = 0.6$ Ω /m, $L' = 38.2$ nH/m, $G' = 0.5$ mS/m, $C' = 23.9$ pF/m
- 2.18

(a) $b = 4.2$ mm
(b) $u_p = 2 \times 10^8$ m/s
- 2.21

$Z_L = (90 - j120)$ Ω
- 2.23

$Z_0 = 55.9$ Ω
- 2.27

$Z_{in} = (40 + j20)$ Ω
- 2.31

(a) $\Gamma = 0.62e^{-j29.7^\circ}$
- 2.32

(b) $\Gamma = 0.16e^{-j80.54^\circ}$
- 2.33

(a) $Z_{in_1} = (35.20 - j8.62)$ Ω
- 2.35

$L = 8.3 \times 10^{-9}$ H
- 2.37

$l = \lambda/4 + n\lambda/2$
- 2.39

$Z_{in} = \frac{100^2}{33.33} = 300$ Ω
- 2.41

(b) $i_L(t) = 3\cos(6\pi \times 10^8t - 135^\circ)$ (A)
- 2.42

(a) $Z_{in} = (41.25 - j16.35)$ Ω
- 2.44

$P_{av}^i = 10.0$ mW; $P_{av}^r = -1.1$ mW; $P_{av}^t = 8.9$ mW
- 2.45

(a) $P_{av} = 0.29$ W
- 2.47

(b) $\Gamma = 0.62e^{-29.7^\circ}$
- 2.50

$Z_{in} = (66 - j125)$ Ω
- 2.52

$Z_{01} = 40$ Ω ; $Z_{02} = 250$ Ω
- 2.53

(b) $S = 1.64$
- 2.55

(a) $Z_{in} = -j154$ Ω
(b) $0.074\lambda + (n\lambda/2)$, $n = 0, 1, 2, \dots$
- 2.57

$y_L = 0.55 + j0.26$
- 2.61

$Z_L = (41 - j19.5)$ Ω

第1章

- 1.1

10 cm
- 1.3

$p(x,t) = 32.36\cos(4\pi \times 10^3t - 12.12\pi x + 36^\circ)$ (N/m²)
- 1.6

$u_p = 0.83$ (m/s); $\lambda = 10.47$ m
- 1.8

(a) $y_1(x,t)$ 沿 +x 方向传播, $y_2(x,t)$ 沿 -x 方向传播。
- 1.10

$\lambda = 1.4$ m
- 1.12

$y_2(t)$ 的相位比 $y_1(t)$ 滞后 30° 。
- 1.14

$\alpha = 2 \times 10^{-3}$ (Np/m)
- 1.17

(c) $z_1z_2 = 18e^{j109.4^\circ}$
- 1.18

(b) $z_2 = 1.22(-1 + j)$
- 1.20

(c) $|z|^2 = 20$
- 1.21

(d) $t = 0$; $s = 6e^{j30^\circ}$
- 1.23

$\ln(z) = 1.76 - j1.03$
- 1.26

$v_c(t) = 15.57\cos(2\pi \times 10^3t - 81.5^\circ)$ V
- 1.27

(d) $i(t) = 3.61\cos(\omega t + 146.31^\circ)$ A
- 1.28

(d) $\tilde{I} = -2e^{j3\pi/4} = 2e^{-j\pi}e^{j3\pi/4} = 2e^{-j\pi/4}$ A

第2章

- 2.1

(a) $l/\lambda = 1.33 \times 10^{-5}$, 可忽略传输线效应;
(c) $l/\lambda = 0.4$, 应考虑传输线效应。
- 2.4

$R' = 1.38$ (Ω /m), $L' = 1.57 \times 10^{-7}$ (H/m), $G' = 0$, $C' = 1.84 \times 10^{-10}$ (F/m)
- 2.8

$\alpha = 0.109$ Np/m; $\beta = 44.5$ rad/m; $Z_0 = (19.6 + j0.030)$ Ω ; $u_p = 1.41 \times 10^8$ m/s
- 2.10

$w = 1.542$ mm, $\lambda = 0.044$ m

- 2.14

$R' = 1$ (Ω /m); $L' = 200$ (nH/m); $G' = 400$ (μ S/m); $C' = 80$ (pF/m); $\lambda = 2.5$ m
- 2.16

$R' = 0.6$ Ω /m, $L' = 38.2$ nH/m, $G' = 0.5$ mS/m, $C' = 23.9$ pF/m
- 2.18

(a) $b = 4.2$ mm
(b) $u_p = 2 \times 10^8$ m/s
- 2.21

$Z_L = (90 - j120)$ Ω
- 2.23

$Z_0 = 55.9$ Ω
- 2.27

$Z_{in} = (40 + j20)$ Ω
- 2.31

(a) $\Gamma = 0.62e^{-j29.7^\circ}$
- 2.32

(b) $\Gamma = 0.16e^{-j80.54^\circ}$
- 2.33

(a) $Z_{in_1} = (35.20 - j8.62)$ Ω
- 2.35

$L = 8.3 \times 10^{-9}$ H
- 2.37

$l = \lambda/4 + n\lambda/2$
- 2.39

$Z_{in} = \frac{100^2}{33.33} = 300$ Ω
- 2.41

(b) $i_L(t) = 3\cos(6\pi \times 10^8t - 135^\circ)$ (A)
- 2.42

(a) $Z_{in} = (41.25 - j16.35)$ Ω
- 2.44

$P_{av}^i = 10.0$ mW; $P_{av}^r = -1.1$ mW; $P_{av}^t = 8.9$ mW
- 2.45

(a) $P_{av} = 0.29$ W
- 2.47

(b) $\Gamma = 0.62e^{-j29.7^\circ}$
- 2.50

$Z_{in} = (66 - j125)$ Ω
- 2.52

$Z_{01} = 40$ Ω ; $Z_{02} = 250$ Ω
- 2.53

(b) $S = 1.64$
- 2.55

(a) $Z_{in} = -j154$ Ω
(b) $0.074\lambda + (n\lambda/2)$, $n = 0, 1, 2, \dots$
- 2.57

$y_L = 0.55 + j0.26$
- 2.61

$Z_L = (41 - j19.5)$ Ω

- 3.63

$Z_{\text{in}} = (95 - j70) \ \Omega$
- 3.69

First solution: Stub at $d = 0.199\lambda$ from antenna and stub length $l = 0.125\lambda$. Second solution: $d = 0.375\lambda$ from antenna and stub length $l = 0.375\lambda$.
- 2.73

$Z_{\text{in}} = 100 \ \Omega$
- 2.78

$V_{\text{g}} = 19.2 \text{ V}; \ R_{\text{g}} = 30 \ \Omega; \ l = 525 \text{ m}$
- 2.82

(a)

$l = 1200 \text{ m}$

(b)

$Z_{\text{L}} = 0$

(c)

$R_{\text{g}} = \left(\frac{1 + \Gamma_{\text{g}}}{1 - \Gamma_{\text{g}}}\right) Z_0 = \left(\frac{1 + 0.25}{1 - 0.25}\right) 50 = 83.3 \ \Omega$

(d)

$V_{\text{g}} = 32 \text{ V}$

Chapter 3

- 3.1

$\hat{\mathbf{a}} = \hat{\mathbf{x}}0.32 + \hat{\mathbf{z}}0.95$
- 3.3

Area = 36
- 3.5

(a)

$A = \sqrt{14}; \ \hat{\mathbf{a}}_A = (\hat{\mathbf{x}} + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}3)/\sqrt{14}$

(e)

$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = 20$

(h)

$(\mathbf{A} \times \hat{\mathbf{y}}) \cdot \hat{\mathbf{z}} = 1$
- 3.9

$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{-\hat{\mathbf{x}} - \hat{\mathbf{y}}\mathbf{y} - \hat{\mathbf{z}}2}{\sqrt{5 + y^2}}$
- 3.11

$\hat{\mathbf{a}} = (\hat{\mathbf{x}}2 - \hat{\mathbf{z}}4)/\sqrt{20}$
- 3.13

$\mathbf{A} = \hat{\mathbf{x}}0.8 + \hat{\mathbf{y}}1.6$
- 3.15

$\hat{\mathbf{c}} = \hat{\mathbf{x}}0.37 + \hat{\mathbf{y}}0.56 + \hat{\mathbf{z}}0.74$
- 3.17

$\mathbf{G} = \pm \left(-\hat{\mathbf{x}}\frac{8}{3} + \hat{\mathbf{y}}\frac{8}{3} + \hat{\mathbf{z}}\frac{4}{3}\right)$
- 3.22

(a)

$P_1 = (2.24, 63.4^\circ, 0)$ in cylindrical;
 $P_1 = (2.24, 90^\circ, 63.4^\circ)$ in spherical

(d)

$P_4 = (2.83, 135^\circ, -2)$ in cylindrical;
 $P_4 = (3.46, 125.3^\circ, 135^\circ)$ in spherical
- 3.24

(a) $P_1 = (0, 0, 5)$
- 3.25

(c) $A = 12$
- 3.26

(a) $V = 21\pi/2$
- 3.30

(a)

$\theta_{AB} = 90^\circ$

(b)

$\pm(\hat{\mathbf{r}}0.487 + \hat{\phi}0.228 + \hat{\mathbf{z}}0.843)$
- 3.32

(a) $d = \sqrt{3}$
- 3.34

(c)

$\vec{C}(P_3) = \hat{\mathbf{r}}0.707 + \hat{\mathbf{z}}4$

(e)

$\mathbf{E}(P_5) = -\hat{\mathbf{r}} + \hat{\phi}$
- 3.35

(c) $\mathbf{C}(P_3) = \hat{\mathbf{R}}0.854 + \hat{\theta}0.146 - \hat{\phi}0.707$
- 3.36

(e) $\nabla S = \hat{\mathbf{x}}8xe^{-z} + \hat{\mathbf{y}}3y^2 - \hat{\mathbf{z}}4x^2e^{-z}$
- 3.37

(b)

$\nabla T = \hat{\mathbf{x}}2x$

(g)

$\nabla T = -\hat{\mathbf{x}}\frac{2\pi}{6}\sin\left(\frac{\pi x}{3}\right)$
- 3.38

$T(z) = 10 + (1 - e^{-3z})/3$
- 3.40

$\left(\frac{dV}{dt}\right)\Big|_{(1,-1,4)} = 2.18$

- 3.43

$dU/dl = -0.02$
- 3.47

$\mathbf{E} = \hat{\mathbf{R}}4R$
- 3.50

(a)

$\oint \mathbf{D} \cdot d\mathbf{s} = 150\pi$

(b)

$\iiint \nabla \cdot \mathbf{D} d\mathcal{V} = 150\pi$
- 3.58

(a)

$\mathbf{A}$ is solenoidal, but not conservative.

(d)

$\mathbf{D}$ is conservative, but not solenoidal.

(h)

$\mathbf{H}$ is conservative, but not solenoidal.
- 3.59

(c) $\nabla^2\left(\frac{3}{x^2 + y^2}\right) = \frac{12}{(x^2 + y^2)^2}$

Chapter 4

- 4.1

$Q = 2.62 \text{ (mC)}$
- 4.3

$Q = 86.65 \text{ (mC)}$
- 4.7

$I = 314.2 \text{ A}$
- 4.9

(a) $\rho_l = -\frac{\pi ca^4}{2} \text{ (C/m)}$
- 4.11

$\mathbf{E} = \hat{\mathbf{z}}51.2 \text{ kV/m}$
- 4.13

$q_2 \approx -94.69 \text{ (}\mu\text{C)}$
- 4.15

(a) $\mathbf{E} = -\hat{\mathbf{x}}1.6 - \hat{\mathbf{y}}0.66 \text{ (MV/m)}$
- 4.17

$\mathbf{E} = \hat{\mathbf{z}}(\rho_{\text{s}0}h/2\epsilon_0)\left[\sqrt{a^2 + h^2} + h^2/\sqrt{a^2 + h^2} - 2h\right]$
- 4.19

$\mathbf{E} = 0$
- 4.23

(a)

$\rho_v = y^3z^3$

(b)

$Q = 32 \text{ (C)}$

(c)

$Q = 32 \text{ (C)}$
- 4.25

$Q = 4\pi\rho_0a^3 \text{ (C)}$
- 4.27

$\mathbf{D} = \hat{\mathbf{r}}\frac{\rho_{\text{v}0}(r^2 - 1)}{2r}, \quad 1 \leq r \leq 3 \text{ m}$

$\mathbf{D} = \hat{\mathbf{r}}D_r = \hat{\mathbf{r}}\frac{4\rho_{\text{v}0}}{r}, \quad r \geq 3 \text{ m}$
- 4.30

$R_1 = \frac{a}{2}, \ R_3 = \frac{a\sqrt{5}}{2}, \ V = \frac{0.55Q}{\pi\epsilon_0a}$
- 4.32

(b) $\mathbf{E} = \hat{\mathbf{z}}(\rho_la/2\epsilon_0)[z/(a^2 + z^2)^{3/2}] \text{ (V/m)}$
- 4.34

$V(b) = \frac{\rho_l}{4\pi\epsilon} \ln\left[\frac{l + \sqrt{l^2 + 4b^2}}{-l + \sqrt{l^2 + 4b^2}}\right] \text{ (V)}$
- 4.37

$V = \frac{\rho_l}{2\pi\epsilon_0}\left[\ln\frac{a}{\sqrt{(x-a)^2 + y^2}}\right] - \ln\frac{a}{\sqrt{(x+a)^2 + y^2}}\Bigg]$
- 4.39

$V_{AB} = -117.09 \text{ V}$
- 4.41

(c) $\mathbf{u}_{\text{e}} = -8.125\mathbf{E}/|\mathbf{E}| \text{ (m/s); } \mathbf{u}_{\text{h}} = 3.125\mathbf{E}/|\mathbf{E}| \text{ (m/s)}$
- 4.45

$R = 4.2 \text{ (m}\Omega\text{)}$
- 4.48

$\theta = 61^\circ$

- 2.63

$Z_{\text{in}} = (95 - j70) \ \Omega$
- 2.69

第一组解：短路支节距天线 $d = 0.199\lambda$ ，支节长度 $l = 0.125\lambda$ 。第二组解： $d = 0.375\lambda$ ， $l = 0.375\lambda$ 。
- 2.73

$Z_{\text{in}} = 100 \ \Omega$
- 2.78

$V_{\text{g}} = 19.2 \text{ V}; \ R_{\text{g}} = 30 \ \Omega; \ l = 525 \text{ m}$
- 2.82

(a)

$l = 1200 \text{ m}$

(b)

$Z_{\text{L}} = 0$

(c)

$R_{\text{g}} = \left(\frac{1 + \Gamma_{\text{g}}}{1 - \Gamma_{\text{g}}}\right) Z_0 = \left(\frac{1 + 0.25}{1 - 0.25}\right) 50 = 83.3 \ \Omega$

(d)

$V_{\text{g}} = 32 \text{ V}$

第3章

- 3.1

$\hat{\mathbf{a}} = \hat{\mathbf{x}}0.32 + \hat{\mathbf{z}}0.95$
- 3.3

面积 = 36
- 3.5

(a)

$A = \sqrt{14}; \ \hat{\mathbf{a}}_A = (\hat{\mathbf{x}} + \hat{\mathbf{y}}2 - \hat{\mathbf{z}}3)/\sqrt{14}$

(e)

$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = 20$

(h)

$(\mathbf{A} \times \hat{\mathbf{y}}) \cdot \hat{\mathbf{z}} = 1$
- 3.9

$\hat{\mathbf{a}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{-\hat{\mathbf{x}} - \hat{\mathbf{y}}\mathbf{y} - \hat{\mathbf{z}}2}{\sqrt{5 + y^2}}$
- 3.11

$\hat{\mathbf{a}} = (\hat{\mathbf{x}}2 - \hat{\mathbf{z}}4)/\sqrt{20}$
- 3.13

$\mathbf{A} = \hat{\mathbf{x}}0.8 + \hat{\mathbf{y}}1.6$
- 3.15

$\hat{\mathbf{c}} = \hat{\mathbf{x}}0.37 + \hat{\mathbf{y}}0.56 + \hat{\mathbf{z}}0.74$
- 3.17

$\mathbf{G} = \pm \left(-\hat{\mathbf{x}}\frac{8}{3} + \hat{\mathbf{y}}\frac{8}{3} + \hat{\mathbf{z}}\frac{4}{3}\right)$
- 3.22

(a) P_1 ：圆柱坐标 $(2.24, 63.4^\circ, 0)$ ；球坐标 $(2.24, 90^\circ, 63.4^\circ)$ 。

(d) P_4 ：圆柱坐标 $(2.83, 135^\circ, -2)$ ；球坐标 $(3.46, 125.3^\circ, 135^\circ)$ 。
- 3.24

(a) $P_1 = (0, 0, 5)$
- 3.25

(c) $A = 12$
- 3.26

(a) $V = 21\pi/2$
- 3.30

(a)

$\theta_{AB} = 90^\circ$

(b)

$\pm(\hat{\mathbf{r}}0.487 + \hat{\phi}0.228 + \hat{\mathbf{z}}0.843)$
- 3.32

(a) $d = \sqrt{3}$
- 3.34

(c)

$\vec{C}(P_3) = \hat{\mathbf{r}}0.707 + \hat{\mathbf{z}}4$

(e)

$\mathbf{E}(P_5) = -\hat{\mathbf{r}} + \hat{\phi}$
- 3.35

(c) $\mathbf{C}(P_3) = \hat{\mathbf{R}}0.854 + \hat{\theta}0.146 - \hat{\phi}0.707$
- 3.36

(e) $\nabla S = \hat{\mathbf{x}}8xe^{-z} + \hat{\mathbf{y}}3y^2 - \hat{\mathbf{z}}4x^2e^{-z}$
- 3.37

(b)

$\nabla T = \hat{\mathbf{x}}2x$

(g)

$\nabla T = -\hat{\mathbf{x}}\frac{2\pi}{6}\sin\left(\frac{\pi x}{3}\right)$
- 3.38

$T(z) = 10 + (1 - e^{-3z})/3$
- 3.40

$\left(\frac{dV}{dt}\right)\Big|_{(1,-1,4)} = 2.18$

- 3.43

$dU/dl = -0.02$
- 3.47

$\mathbf{E} = \hat{\mathbf{R}}4R$
- 3.50

(a)

$\oint \mathbf{D} \cdot d\mathbf{s} = 150\pi$

(b)

$\iiint \nabla \cdot \mathbf{D} d\mathcal{V} = 150\pi$
- 3.58

(a) $\mathbf{A}$ 为无散场，但非保守场。

(d) $\mathbf{D}$ 为保守场，但非无散场。

(h) $\mathbf{H}$ 为保守场，但非无散场。
- 3.59

(c) $\nabla^2\left(\frac{3}{x^2 + y^2}\right) = \frac{12}{(x^2 + y^2)^2}$

第4章

- 4.1

$Q = 2.62 \text{ (mC)}$
- 4.3

$Q = 86.65 \text{ (mC)}$
- 4.7

$I = 314.2 \text{ A}$
- 4.9

(a) $\rho_l = -\frac{\pi ca^4}{2} \text{ (C/m)}$
- 4.11

$\mathbf{E} = \hat{\mathbf{z}}51.2 \text{ kV/m}$
- 4.13

$q_2 \approx -94.69 \text{ (}\mu\text{C)}$
- 4.15

(a) $\mathbf{E} = -\hat{\mathbf{x}}1.6 - \hat{\mathbf{y}}0.66 \text{ (MV/m)}$
- 4.17

$\mathbf{E} = \hat{\mathbf{z}}(\rho_{\text{s}0}h/2\epsilon_0)\left[\sqrt{a^2 + h^2} + h^2/\sqrt{a^2 + h^2} - 2h\right]$
- 4.19

$\mathbf{E} = 0$
- 4.23

(a)

$\rho_v = y^3z^3$

(b)

$Q = 32 \text{ (C)}$

(c)

$Q = 32 \text{ (C)}$
- 4.25

$Q = 4\pi\rho_0a^3 \text{ (C)}$
- 4.27

$\mathbf{D} = \hat{\mathbf{r}}\frac{\rho_{\text{v}0}(r^2 - 1)}{2r}, \quad 1 \leq r \leq 3 \text{ m}$

$\mathbf{D} = \hat{\mathbf{r}}D_r = \hat{\mathbf{r}}\frac{4\rho_{\text{v}0}}{r}, \quad r \geq 3 \text{ m}$
- 4.30

$R_1 = \frac{a}{2}, \ R_3 = \frac{a\sqrt{5}}{2}, \ V = \frac{0.55Q}{\pi\epsilon_0a}$
- 4.32

(b) $\mathbf{E} = \hat{\mathbf{z}}(\rho_la/2\epsilon_0)[z/(a^2 + z^2)^{3/2}] \text{ (V/m)}$
- 4.34

$V(b) = \frac{\rho_l}{4\pi\epsilon} \ln\left[\frac{l + \sqrt{l^2 + 4b^2}}{-l + \sqrt{l^2 + 4b^2}}\right] \text{ (V)}$
- 4.37

$V = \frac{\rho_l}{2\pi\epsilon_0}\left[\ln\frac{a}{\sqrt{(x-a)^2 + y^2}}\right] - \ln\frac{a}{\sqrt{(x+a)^2 + y^2}}\Bigg]$
- 4.39

$V_{AB} = -117.09 \text{ V}$
- 4.41

(c) $\mathbf{u}_{\text{e}} = -8.125\mathbf{E}/|\mathbf{E}| \text{ (m/s); } \mathbf{u}_{\text{h}} = 3.125\mathbf{E}/|\mathbf{E}| \text{ (m/s)}$
- 4.45

$R = 4.2 \text{ (m}\Omega\text{)}$
- 4.48

$\theta = 61^\circ$

4.50 $Q = \frac{3\pi\epsilon_0}{2}$ (C)

4.53 **(a)** $|E|$ is maximum at $r = a$.

4.55 $W_e = 4.62 \times 10^{-9}$ (J)

4.57 **(a)** $C = 3.1$ pF

4.60 **(b)** $C = 6.07$ pF

4.63 $C' = \frac{\pi\epsilon_0}{\ln[(2d/a) - 1]}$ (C/m)

Chapter 5

5.1 $\mathbf{a} = -\hat{\mathbf{y}}4.22 \times 10^{18}$ (m/s²)

5.4 $\vec{T} = -\hat{\mathbf{z}}1.66$ (N·m); clockwise

5.5 **(a)** $\mathbf{F} = 0$

5.7 $\mathbf{B} = -\hat{\mathbf{z}}0.6$ (mT)

5.9 $\mathbf{H} = \hat{\mathbf{z}}\frac{I\theta(b-a)}{4\pi ab}$

5.11 $I_2 = \frac{2aI_1}{2\pi Nd} = \frac{1 \times 25}{\pi \times 20 \times 2} = 0.2$ A

5.13 $I = 200$ A

5.16 $\vec{F} = -\hat{\mathbf{x}}0.4$ (mN)

5.18 **(a)** $\mathbf{H}(0, 0, h) = -\hat{\mathbf{x}}\frac{I}{\pi w}\tan^{-1}\left(\frac{w}{2h}\right)$ (A/m)

5.20 $\mathbf{F} = \hat{\mathbf{y}}4 \times 10^{-5}$ N

5.24 $\mathbf{J} = \hat{\mathbf{z}}36e^{-3r}$ A/m²

5.26 **(a)** $\vec{A} = \hat{\mathbf{z}}\frac{\mu_0 I}{4\pi}\ln\left(\frac{\ell + \sqrt{\ell^2 + 4r^2}}{-\ell + \sqrt{\ell^2 + 4r^2}}\right)$

5.27 **(a)** $\mathbf{B} = \hat{\mathbf{z}}5\pi\sin\pi y - \hat{\mathbf{y}}\pi\cos\pi x$ (T)

5.29 **(a)** $\mathbf{A} = \hat{\mathbf{z}}\mu_0 IL/(4/\pi R)$

(b) $\mathbf{H} = (IL/4\pi)[(-\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2 + z^2)^{3/2}]$

5.31 $n_e = 1.5$ electrons/atom

5.33 $\mathbf{H}_2 = \hat{\mathbf{z}}3$

5.35 $\vec{B}_2 = \hat{\mathbf{x}}20,000 - \hat{\mathbf{y}}30,000 + \hat{\mathbf{z}}8$

5.37 $L' = (\mu/\pi)\ln[(d-a)/a]$ (H)

5.40 $\Phi = 1.66 \times 10^{-6}$ (Wb)

Chapter 6

6.1 At $t = 0$, current in top loop is momentarily clockwise. At $t = t_1$, current in top loop is momentarily counterclockwise.

6.3 **(a)** $V_{\text{emf}} = 375e^{-3t}$ (V)

6.5 $B_0 = 0.8$ (nA/m)

6.7 $I_{\text{ind}} = 37.7\sin(200\pi t)$ mA

6.10 $V_{12} = -236$ (μV)

6.12 $I = 0.1$ (A)

6.15 $I = 0.82\cos(120\pi t)$ (μA)

6.17 $f = 5$ MHz

6.18 **(b)** 888

6.20 $\rho_v = (8y/\omega)\sin\omega t + C_0$, where C_0 is a constant of integration.

6.24 $k = (4\pi/30)$ rad/m;
 $\mathbf{E} = -\hat{\mathbf{z}}941\cos(2\pi \times 10^7 t + 4\pi y/30)$ (V/m)

6.26 $\mathbf{H}(R, \theta; t) = \hat{\boldsymbol{\phi}}(53/R)\sin\theta\cos(6\pi \times 10^8 t - 2\pi R)$ ($\mu\text{A/m}$)

6.28 **(a)** $k = 20$ (rad/m)

Chapter 7

7.1 **(a)** Positive y direction

(c) $\lambda = 12.6$ m

7.3 **(a)** $\lambda = 31.42$ m

7.5 $\epsilon_r = 9$

7.7 **(a)** $\lambda = 20$ m

7.9 $\mathbf{E} = \hat{\mathbf{x}}\sqrt{2}\cos(\omega t + kz) - \hat{\mathbf{y}}\sqrt{2}\sin(\omega t + kz)$ (V/m)

7.12 At $x = 0$ and $t = 0$, $\mathbf{E} = -\hat{\mathbf{z}}1.885$ (V/m)

At $x = 0$ and $t = 5$ ns, $\omega t = 0.13$ rad and
 $\mathbf{E} = -1.885(\hat{\mathbf{y}}0.13 + \hat{\mathbf{z}}0.99)$ (V/m)

7.14 **(a)** $\gamma = 73.5^\circ$ and $\chi = -8.73^\circ$

(b) Right-hand elliptically polarized

7.17 **(a)** Low-loss dielectric. $\alpha = 8.42 \times 10^{-11}$ Np/m,
 $\beta = 468.3$ rad/m, $\lambda = 1.34$ cm, $u_p = 1.34 \times 10^8$ m/s,
 $\eta_c \approx 168.5\ \Omega$

7.19 $\mathbf{H}$ lags $\mathbf{E}$ by 31.72°

7.21 $z = 287.82$ m

7.23 $u_p = 9.42 \times 10^4$ (m/s)

7.25 $\mathbf{H} = -\hat{\mathbf{y}}0.16e^{-30x}\cos(2\pi \times 10^9 t - 40x - 36.85^\circ)$ (A/m)

7.29 $(R_{\text{ac}}/R_{\text{dc}}) = 143.55$

7.33 $S_{\text{av}} = \hat{\mathbf{y}}0.48$ (W/m²)

7.35 **(c)** $z = 23.03$ m

7.37 $u_p = 1 \times 10^8$ (m/s)

7.39 **(b)** $P_{\text{av}} = 0$

7.41 **(a)** $(w_e)_{\text{av}} = \frac{\epsilon E_0^2}{4}$

附录 E 部分习题答案 464

4.50 $Q = \frac{3\pi\epsilon_0}{2}$ (C)

4.53 (a) $|E|$ 在 $r = a$ 处达到最大值。

4.55 $W_e = 4.62 \times 10^{-9}$ (J)

4.57 **(a)** $C = 3.1$ pF

4.60 **(b)** $C = 6.07$ pF

4.63 $C' = \frac{\pi\epsilon_0}{\ln[(2d/a) - 1]}$ (C/m)

第5章

5.1 $\mathbf{a} = -\hat{\mathbf{y}}4.22 \times 10^{18}$ (m/s²)

5.4 $\vec{T} = -\hat{\mathbf{z}}1.66$ (N·m); (顺时针)

5.5 **(a)** $\mathbf{F} = 0$

5.7 $\mathbf{B} = -\hat{\mathbf{z}}0.6$ (mT)

5.9 $\mathbf{H} = \hat{\mathbf{z}}\frac{I\theta(b-a)}{4\pi ab}$

5.11 $I_2 = \frac{2aI_1}{2\pi Nd} = \frac{1 \times 25}{\pi \times 20 \times 2} = 0.2$ A

5.13 $I = 200$ A

5.16 $\vec{F} = -\hat{\mathbf{x}}0.4$ (mN)

5.18 **(a)** $\mathbf{H}(0, 0, h) = -\hat{\mathbf{x}}\frac{I}{\pi w}\tan^{-1}\left(\frac{w}{2h}\right)$ (A/m)

5.20 $\mathbf{F} = \hat{\mathbf{y}}4 \times 10^{-5}$ N

5.24 $\mathbf{J} = \hat{\mathbf{z}}36e^{-3r}$ A/m²

5.26 **(a)** $\vec{A} = \hat{\mathbf{z}}\frac{\mu_0 I}{4\pi}\ln\left(\frac{\ell + \sqrt{\ell^2 + 4r^2}}{-\ell + \sqrt{\ell^2 + 4r^2}}\right)$

5.27 **(a)** $\mathbf{B} = \hat{\mathbf{z}}5\pi\sin\pi y - \hat{\mathbf{y}}\pi\cos\pi x$ (T)

5.29 (a) $A_z = \mu_0 IL/(4\pi R)$, 其余分量为零。

(b) $\mathbf{H} = (IL/4\pi)[(-\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2 + z^2)^{3/2}]$

5.31 $n_e = 1.5$ 个电子/原子

5.33 $\mathbf{H}_2 = \hat{\mathbf{z}}3$

5.35 $\vec{B}_2 = \hat{\mathbf{x}}20,000 - \hat{\mathbf{y}}30,000 + \hat{\mathbf{z}}8$

5.37 $L' = (\mu/\pi)\ln[(d-a)/a]$ (H)

5.40 $\Phi = 1.66 \times 10^{-6}$ (Wb)

第6章

6.1 $t = 0$ 时上方回路电流沿顺时针方向; $t = t_1$ 时沿逆时针方向。

6.3 **(a)** $V_{\text{emf}} = 375e^{-3t}$ (V)

6.5 $B_0 = 0.8$ (nA/m)

6.7 $I_{\text{ind}} = 37.7\sin(200\pi t)$ mA

6.10 $V_{12} = -236$ (μV)

6.12 $I = 0.1$ (A)

6.15 $I = 0.82\cos(120\pi t)$ (μA)

6.17 $f = 5$ MHz

6.18 **(b)** 888

6.20 $\rho_v = (8y/\omega)\sin\omega t + C_0$, C_0 为积分常数。

6.24 $k = (4\pi/30)$ rad/m;
 $\mathbf{E} = -\hat{\mathbf{z}}941\cos(2\pi \times 10^7 t + 4\pi y/30)$ (V/m)

6.26 $\mathbf{H}(R, \theta; t) = \hat{\boldsymbol{\phi}}(53/R)\sin\theta\cos(6\pi \times 10^8 t - 2\pi R)$ ($\mu\text{A/m}$)

6.28 **(a)** $k = 20$ (rad/m)

第7章

7.1 (a) 沿 +y 方向

(c) $\lambda = 12.6$ m

7.3 **(a)** $\lambda = 31.42$ m

7.5 $\epsilon_r = 9$

7.7 **(a)** $\lambda = 20$ m

7.9 $\mathbf{E} = \hat{\mathbf{x}}\sqrt{2}\cos(\omega t + kz) - \hat{\mathbf{y}}\sqrt{2}\sin(\omega t + kz)$ (V/m)

7.12 当 $x = 0$ 且 $t = 0$, $\mathbf{E} = -\hat{\mathbf{z}}1.885$ (V/m)

当 $x = 0$ 且 $t = 5$ ns, $\omega t = 0.13$ rad 且
 $\mathbf{E} = -1.885(\hat{\mathbf{y}}0.13 + \hat{\mathbf{z}}0.99)$ (V/m)

7.14 **(a)** $\gamma = 73.5^\circ$ and $\chi = -8.73^\circ$

(b) 右旋椭圆极化

7.17 **(a)** 低损耗电介质; $\alpha = 8.42 \times 10^{-11}$ Np/m,
 $\beta = 468.3$ rad/m, $\lambda = 1.34$ cm, $u_p = 1.34 \times 10^8$ m/s,
 $\eta_c \approx 168.5\ \Omega$

7.19 H 的相位比 E 滞后 31.72° 。

7.21 $z = 287.82$ m

7.23 $u_p = 9.42 \times 10^4$ (m/s)

7.25 $\mathbf{H} = -\hat{\mathbf{y}}0.16e^{-30x}\cos(2\pi \times 10^9 t - 40x - 36.85^\circ)$ (A/m)

7.29 $(R_{\text{ac}}/R_{\text{dc}}) = 143.55$

7.33 $S_{\text{av}} = \hat{\mathbf{y}}0.48$ (W/m²)

7.35 **(c)** $z = 23.03$ m

7.37 $u_p = 1 \times 10^8$ (m/s)

7.39 **(b)** $P_{\text{av}} = 0$

7.41 **(a)** $(w_e)_{\text{av}} = \frac{\epsilon E_0^2}{4}$

校注：5.29 按短电流元的磁矢势公式修正原书分母中的 “/pi” 误印。

Chapter 8

- 8.1 (a) $\Gamma = -0.67$; $\tau = 0.33$
 (b) $S = 5$
 (c) $S_{\text{av}}^{\text{i}} = 0.52 \text{ (W/m}^2\text{)}$; $S_{\text{av}}^{\text{r}} = 0.24 \text{ (W/m}^2\text{)}$;
 $S_{\text{av}}^{\text{t}} = 0.28 \text{ (W/m}^2\text{)}$
- 8.3 (b) $S_{\text{av}}^{\text{i}} = \hat{y} 251.34$, $S_{\text{av}}^{\text{r}} = \hat{y} 10.05$,
 $S_{\text{av}}^{\text{t}} = \hat{y} 241.29 \quad \text{(W/m}^2\text{)}$
- 8.6 (a) $\Gamma = -0.71$
- 8.7 $|\tilde{\mathbf{E}}_1|_{\text{max}} = 85.5 \text{ (V/m)}$; $l_{\text{max}} = 1.5 \text{ m}$
- 8.9 $\epsilon_{\text{r}_2} = \sqrt{\epsilon_{\text{r}_1} \epsilon_{\text{r}_3}}$; $d = c/[4f(\epsilon_{\text{r}_1} \epsilon_{\text{r}_3})^{1/4}]$
- 8.11 $Z_{\text{in}}(-d) = 0.43\eta_0/\underline{-51.7^\circ}$
 $|\Gamma|^2 = 0.24$
- 8.13 $f = 75 \text{ MHz}$
- 8.15 $P' = (3.3 \times 10^{-3})^2 \frac{10^2}{2} \times 1.14 [1 - e^{-2 \times 44.43 \times 2 \times 10^{-3}}]$
 $= 1.01 \times 10^{-4} \quad \text{(W/m}^2\text{)}$
- 8.17 $\theta_{\text{min}} = 20.4^\circ$
- 8.19 $\frac{S^{\text{t}}}{S^{\text{i}}} = 0.85$
- 8.22 $d = 15 \text{ cm}$
- 8.24 $d = 68.42 \text{ cm}$
- 8.26 $f_{\text{p}} = 59.88 \text{ (Mb/s)}$
- 8.29 (b) $\theta_{\text{t}} = 36.87^\circ$
- 8.31 (a) $\theta_{\text{t}} = 33.7^\circ$
- 8.33 $\theta_{\text{t}} = 18.44^\circ$
- 8.39 (a) $R = 6.4 \times 10^{-3}$; $T = 0.9936$
 (b) $P^{\text{i}} = 85 \text{ mW}$; $P^{\text{r}} = 0.55 \text{ mW}$; $P^{\text{t}} = 84.45 \text{ mW}$
- 8.42 (a) 9.4%
- 8.47 $a = 3.33 \text{ cm}$; $b = 2 \text{ cm}$
- 8.49 Any one of the first four modes.
- 8.51 $570 \text{ } \Omega$ (empty); $290 \text{ } \Omega$ (filled)
- 8.53 $\theta'_{20} = 57.7^\circ$
- 8.55 (a) $Q = 8367$

Chapter 9

- 9.1 $S_{\text{max}} = 7.6 \text{ } (\mu\text{W/m}^2)$
- 9.4 (a) Direction of maximum radiation is a circular cone 120°
 wide, centered around the $+z$ axis.
 (b) $D = 4 = 6 \text{ dB}$
 (c) $\Omega_{\text{p}} = \pi \text{ (sr)} = 3.14 \text{ (sr)}$
 (d) $\beta = 120^\circ$
- 9.6 (b) $G = -3.5 \text{ dB}$

第8章

- 8.1 (a) $\Gamma = -0.67$; $\tau = 0.33$
 (b) $S = 5$
 (c) $S_{\text{av}}^{\text{i}} = 0.52 \text{ (W/m}^2\text{)}$; $S_{\text{av}}^{\text{r}} = 0.24 \text{ (W/m}^2\text{)}$;
 $S_{\text{av}}^{\text{t}} = 0.28 \text{ (W/m}^2\text{)}$
- 8.3 (b) $S_{\text{av}}^{\text{i}} = \hat{y} 251.34$, $S_{\text{av}}^{\text{r}} = \hat{y} 10.05$,
 $S_{\text{av}}^{\text{t}} = \hat{y} 241.29 \quad \text{(W/m}^2\text{)}$
- 8.6 (a) $\Gamma = -0.71$
- 8.7 $|\tilde{\mathbf{E}}_1|_{\text{max}} = 85.5 \text{ (V/m)}$; $l_{\text{max}} = 1.5 \text{ m}$
- 8.9 $\epsilon_{\text{r}_2} = \sqrt{\epsilon_{\text{r}_1} \epsilon_{\text{r}_3}}$; $d = c/[4f(\epsilon_{\text{r}_1} \epsilon_{\text{r}_3})^{1/4}]$
- 8.11 $Z_{\text{in}}(-d) = 0.43\eta_0/\underline{-51.7^\circ}$
 $|\Gamma|^2 = 0.24$
- 8.13 $f = 75 \text{ MHz}$
- 8.15 $P' = (3.3 \times 10^{-3})^2 \frac{10^2}{2} \times 1.14 [1 - e^{-2 \times 44.43 \times 2 \times 10^{-3}}]$
 $= 1.01 \times 10^{-4} \quad \text{(W/m}^2\text{)}$
- 8.17 $\theta_{\text{min}} = 20.4^\circ$
- 8.19 $\frac{S^{\text{t}}}{S^{\text{i}}} = 0.85$
- 8.22 $d = 15 \text{ cm}$
- 8.24 $d = 68.42 \text{ cm}$
- 8.26 $f_{\text{p}} = 59.88 \text{ (Mb/s)}$
- 8.29 (b) $\theta_{\text{t}} = 36.87^\circ$
- 8.31 (a) $\theta_{\text{t}} = 33.7^\circ$
- 8.33 $\theta_{\text{t}} = 18.44^\circ$
- 8.39 (a) $R = 6.4 \times 10^{-3}$; $T = 0.9936$
 (b) $P^{\text{i}} = 85 \text{ mW}$; $P^{\text{r}} = 0.55 \text{ mW}$; $P^{\text{t}} = 84.45 \text{ mW}$
- 8.42 (a) 9.4%
- 8.47 $a = 3.33 \text{ cm}$; $b = 2 \text{ cm}$
- 8.49 前四个模式中的任意一个。
- 8.51 空腔： $570 \text{ } \Omega$ ； 填充介质后： $290 \text{ } \Omega$ 。
- 8.53 $\theta'_{20} = 57.7^\circ$
- 8.55 (a) $Q = 8367$

第9章

- 9.1 $S_{\text{max}} = 7.6 \text{ } (\mu\text{W/m}^2)$
- 9.4 (a) 最大辐射方向构成绕 $+z$ 轴的圆锥面，圆锥全顶角
 为 120° 。
 (b) $D = 4 = 6 \text{ dB}$
 (c) $\Omega_{\text{p}} = \pi \text{ (sr)} = 3.14 \text{ (sr)}$
 (d) $\beta = 120^\circ$
- 9.6 (b) $G = -3.5 \text{ dB}$

- 9.9 $S_{\text{max}} = 4 \times 10^{-5} \text{ (W/m}^2\text{)}$
- 9.11 $D = 36.61 \text{ dB}$
- 9.14 $S = 1.46$
- 9.16 (a) $\tilde{\mathbf{E}}(R, \theta, \phi) = \hat{\boldsymbol{\theta}} \tilde{E}_\theta = \hat{\boldsymbol{\theta}} j \frac{I_0 l k \eta_0}{8\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta \text{ (V/m)}$
- 9.17 (a) $\theta_{\text{max}_1} = 42.6^\circ$, $\theta_{\text{max}_2} = 137.4^\circ$
- 9.20 (a) $\theta_{\text{max}_1} = 90^\circ$, $\theta_{\text{max}_2} = 270^\circ$
 (b) $S_{\text{max}} = \frac{60I_0^2}{\pi R^2}$
 (c) $F(\theta) = \frac{1}{4} \left[\frac{\cos(\pi \cos \theta) + 1}{\sin \theta} \right]^2$
- 9.26 $P_{\text{t}} = 259 \text{ (mW)}$
- 9.28 $P_{\text{t}} = 75 \text{ } (\mu\text{W)}$
- 9.30 (a) $P_{\text{rec}} = 3.6 \times 10^{-6} \text{ W}$
- 9.33 $\beta_{\text{null}} = 5.73^\circ$
- 9.35 $D = 45.6 \text{ dB}$
- 9.38 (a) $\beta_{\text{e}} = 1.8^\circ$; $\beta_{\text{a}} = 0.18^\circ$
 (b) $\Delta y = \beta_{\text{a}} R = 0.96 \text{ m}$
- 9.42 (a) $F_{\text{a}}(\theta) = 4\cos^2 \left[\frac{\pi}{8} (4\cos \theta + 1) \right]$
- 9.44 $d/\lambda = 1.414$
- 9.49 $F_a(\theta) = [6 + 8\cos(\pi \cos \theta) + 2\cos(2\pi \cos \theta)]^2$
- 9.51 $\delta = -2.72 \text{ (rad)} = -155.9^\circ$
- 第10章
- 10.1 T = 107.26 min
- 10.3 133.3, 即约 133 个信道。
- 10.5 $(f_{\text{p}})_{\text{max}} = 300 \text{ kHz}$
- 10.7 $R_{\text{max}} = 4.84 \text{ km}$